Molecular Dynamics:

From Learning the Fundamentals to a Complete Water Simulation

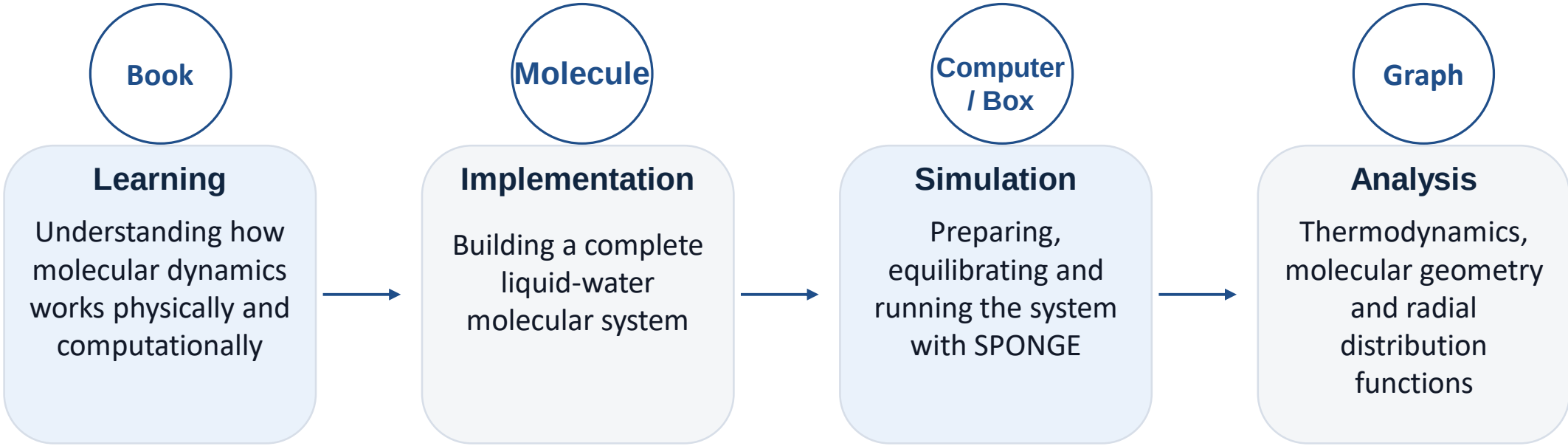
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Under the supervision
of
Prof. Yiqin Gao and Dr. Manas Mondal

My Journey



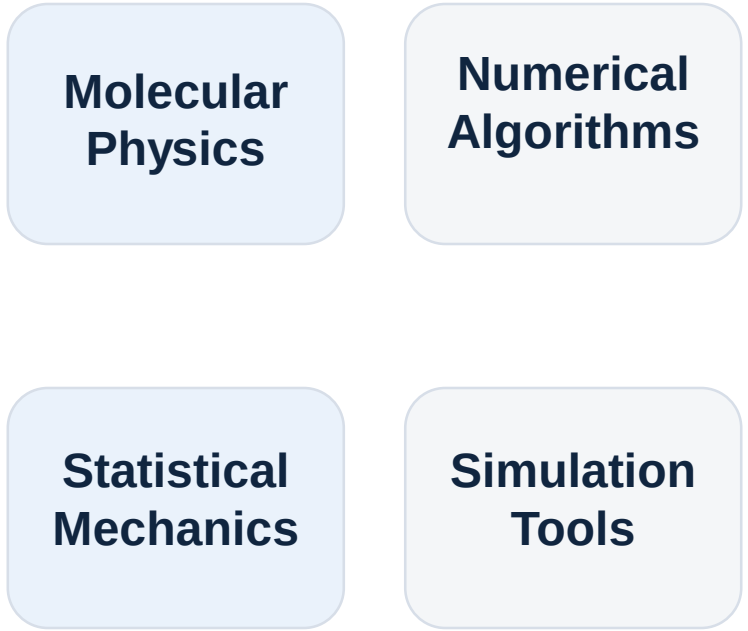
Learn → Build → Simulate → Analyze

What I First Needed to Learn

Before running simulations, I first studied four main areas:

- Molecular interactions and force calculation
- Numerical algorithms for atomic motion
- Periodic systems, electrostatics and thermodynamic ensembles
- Practical MD tools and the connection between MD and AI

These were exactly the four broad learning areas developed from the mentor-provided materials.



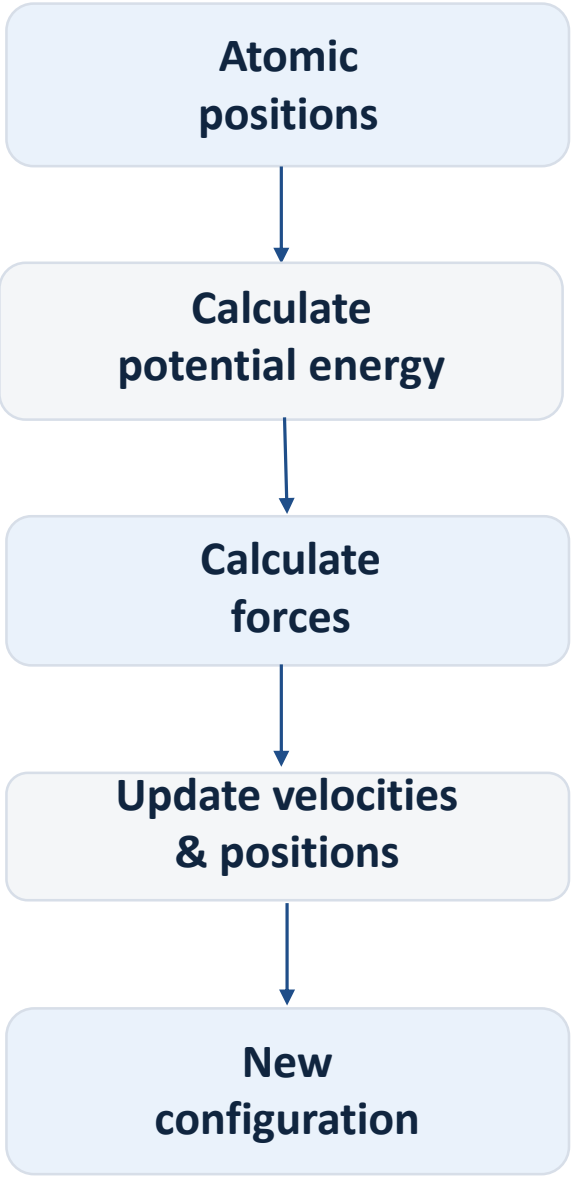
What Is Molecular Dynamics?

Molecular Dynamics (MD) simulates how atoms move through time by repeatedly applying Newton's laws.

$$F_i = -\nabla_i U$$

$$a_i = F_i / m_i$$

Then a numerical integration algorithm updates the positions and velocities.

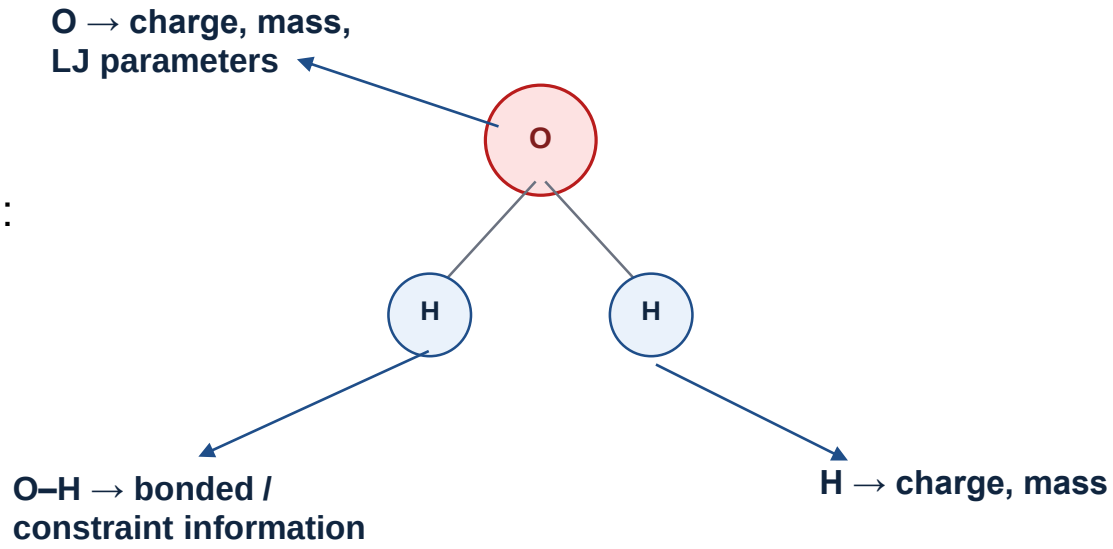


How a Molecular System Is Represented

A simulation needs more than atomic coordinates.

For each atom or molecule we need information such as:

- position
- velocity
- mass
- charge
- atom type
- molecular connectivity
- interaction parameters



Molecular structure → numerical data used by the MD engine

Force Fields: Turning Chemistry Into Equations

A classical force field approximates the potential energy:

$$U = U_{bonded} + U_{nonbonded}$$

Conceptually:

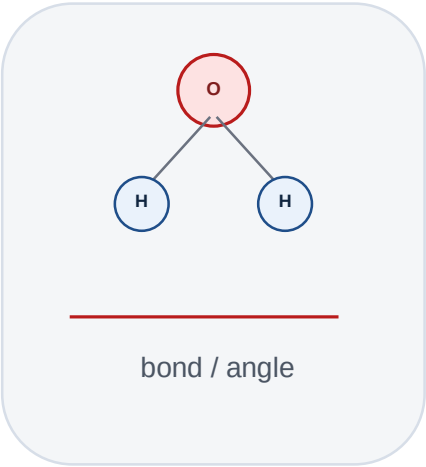
Bonded interactions

- bonds
- angles
- dihedrals

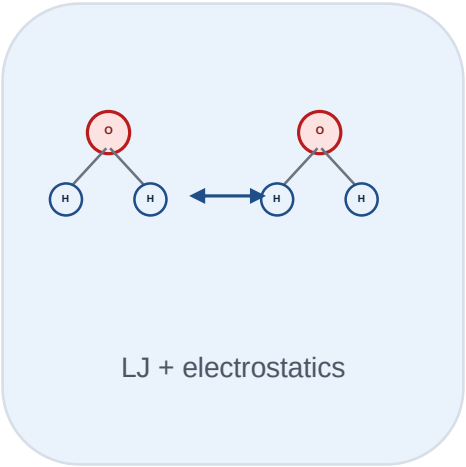
Non-bonded interactions

- van der Waals / Lennard-Jones
- electrostatics

Bonded



Non-bonded



The force is obtained from the gradient of this potential.

From Forces to Motion

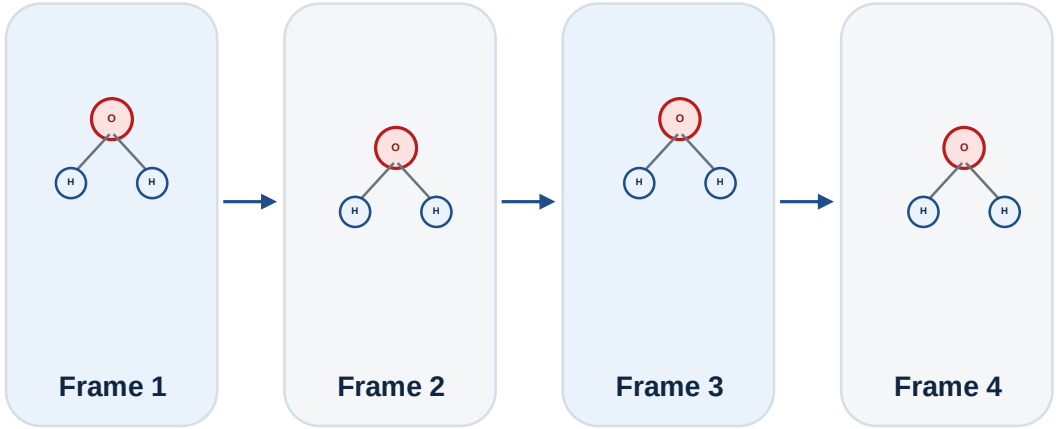
The computer cannot solve molecular motion continuously.

Instead:

$$t,\ t + \Delta t,\ t + 2\Delta t,\ \ldots$$

At every small timestep:

1. update motion;
2. calculate new forces;
3. update motion again;
4. repeat.



Frame 1 → Frame 2 → Frame 3 → Frame 4

Accuracy ↔ computational cost

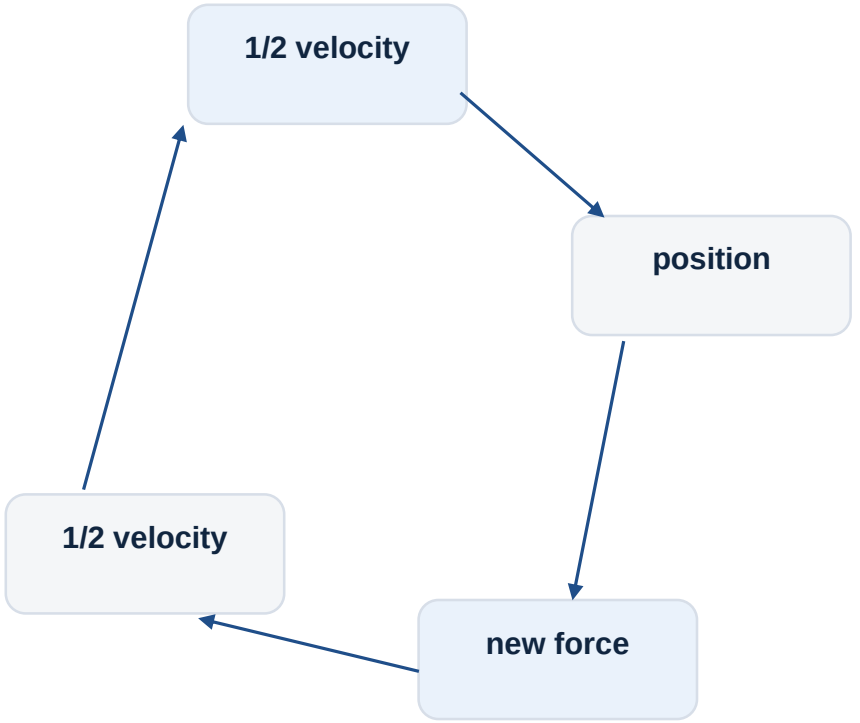
Smaller timestep → more stable/accurate
Larger timestep → faster but potentially unstable

Numerical Integration and Velocity Verlet

One important algorithm I studied was Velocity Verlet.

Conceptually:

Half velocity update
↓
Position update
↓
New force calculation
↓
Second half velocity update
↓
repeat



Velocity Verlet loop

The earlier learning work specifically covered Verlet, Velocity Verlet and Leapfrog, together with timestep stability and numerical error.

Preparing a System Before MD

A simulation cannot normally start directly from arbitrary coordinates.

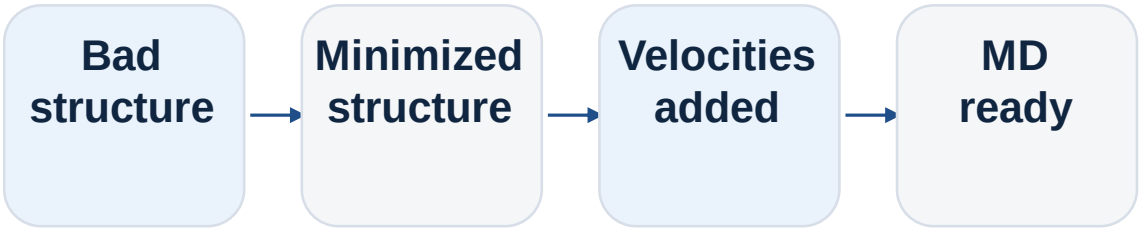
Preparation includes:

Initial positions
→ avoid overlaps and unrealistic geometries

Energy minimization
→ move toward a physically reasonable low-energy structure

Initial velocities
→ assign a temperature-dependent velocity distribution

Constraints
→ keep selected molecular geometry fixed



Energy minimization is preparation — it is not physical simulation time.

The learning phase covered initial position preparation, velocity assignment, minimization and constraints including SHAKE, RATTLE, LINCS and SETTLE.

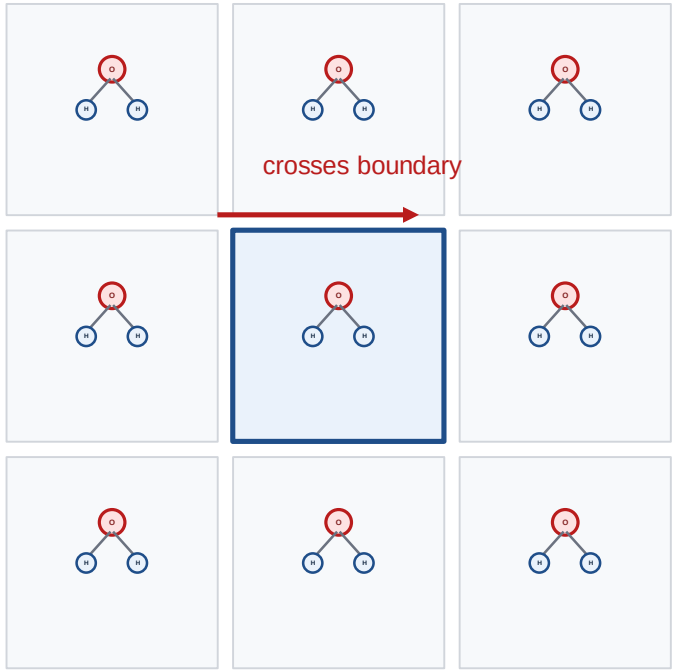
Periodic Boundary Conditions

A small simulation box has artificial surfaces.

To approximate bulk material, the box is repeated periodically.

When a molecule leaves one side:

it enters through the opposite side.



3 × 3 periodic copies

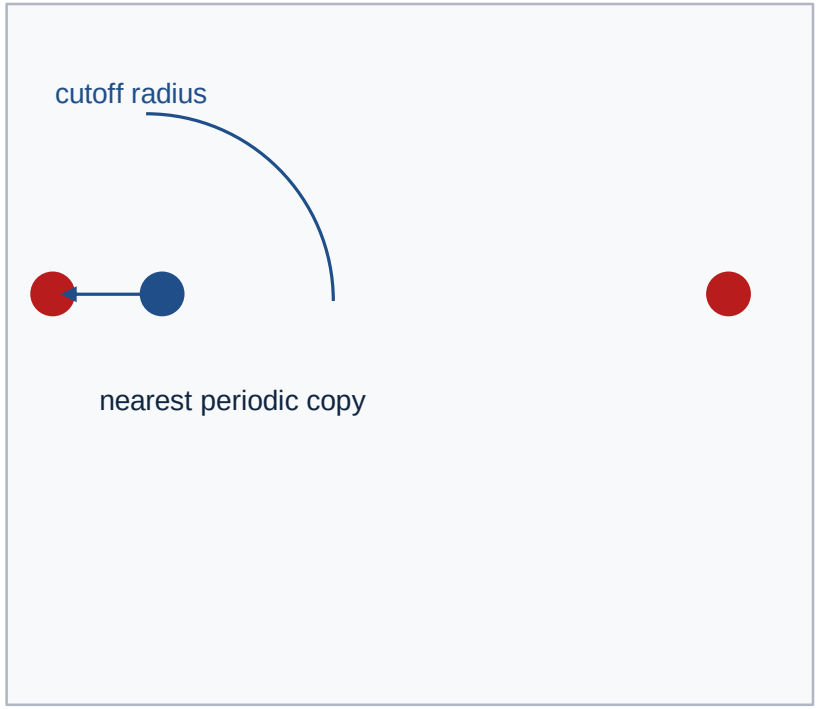
Minimum Image + Non-Bonded Cutoff

For short-range interactions:

use the nearest periodic copy of another atom.

A cutoff reduces computational cost by ignoring sufficiently distant short-range interactions.

But electrostatics cannot simply be truncated in the same way because Coulomb interactions are long-ranged.



Central box

Long-Range Electrostatics: PME

Particle Mesh Ewald

PME separates electrostatic calculations into:

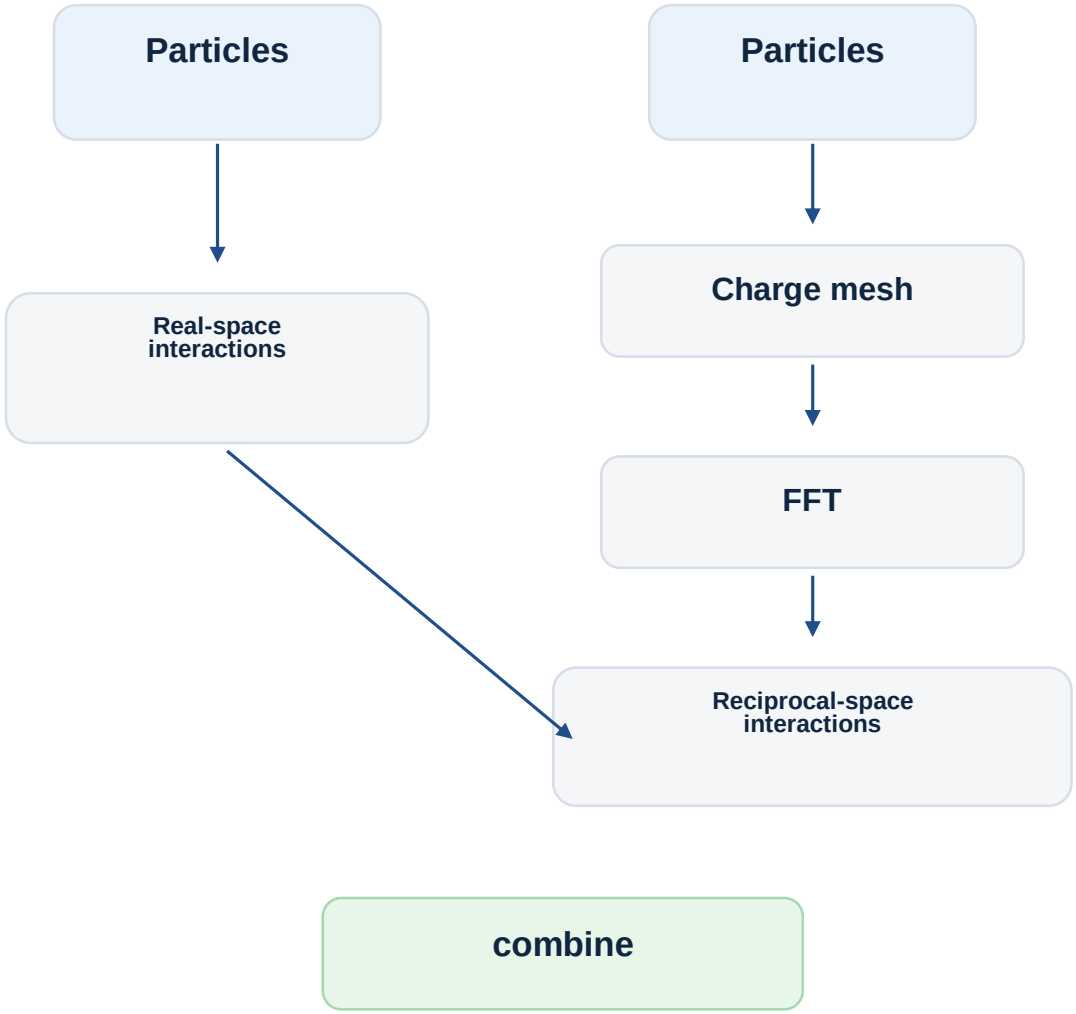
Short-range

→ direct calculation in real space

Long-range

→ charges mapped to a mesh
→ FFT
→ reciprocal-space contribution

Then both contributions are combined.

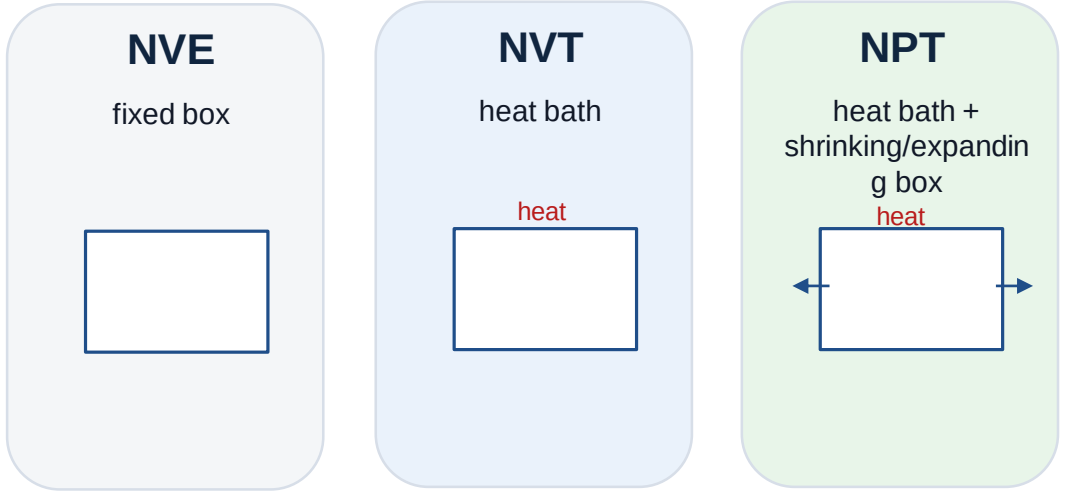


NVE, NVT and NPT Ensembles

Ensemble	Fixed/controlled	Main use
NVE	N, V, E	Numerical/stability studies
NVT	N, V, T	Temperature equilibration
NPT	N, P, T	Density/volume equilibration and pressure-controlled simulation

Thermostat controls temperature.

Barostat controls pressure/volume.



Thermostats and Barostats

I studied several thermostat approaches:

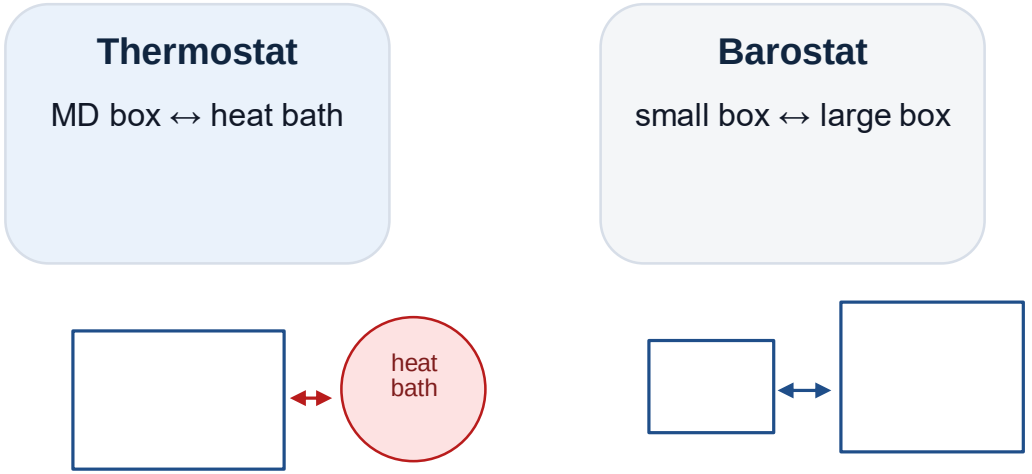
- velocity rescaling
- Andersen
- Berendsen
- Langevin
- Nosé–Hoover

They differ in:

- how temperature is controlled;
- how fluctuations are reproduced;
- how much dynamics is disturbed.

For pressure:

barostat → changes box dimensions / volume



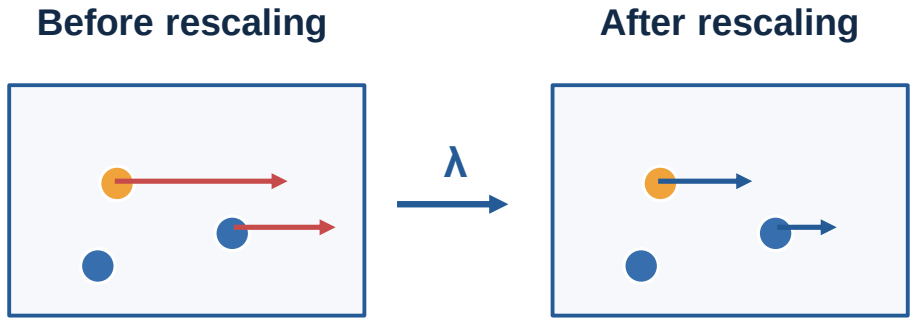
1. Velocity Rescaling

Core idea

Multiply every particle velocity by the same factor so the instantaneous temperature moves to the target.

$$\lambda = \sqrt{(T_{target} / T_{inst})} \quad \text{and} \quad v_i \leftarrow \lambda v_i$$

If T is too high → shorten all velocity arrows
If T is too low → lengthen all velocity arrows



Strengths

- Very simple
- Rapidly reaches target T

Limitations

- Basic form does not reproduce canonical temperature fluctuations
- Abrupt scaling can disturb dynamics

Best for initialization or simple temperature correction — not the best choice for rigorous production sampling.

2. Andersen Thermostat

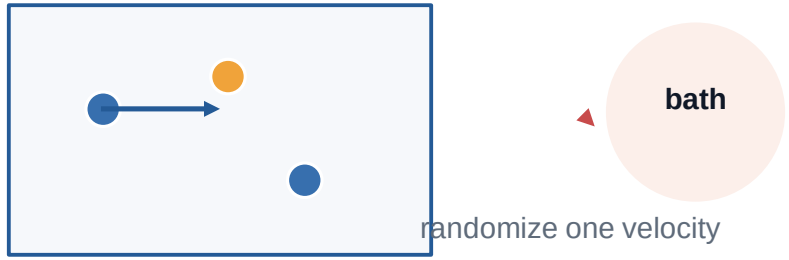
Core idea

Particles experience random “collisions” with an imaginary heat bath. When a collision occurs, the selected particle receives a new velocity drawn from the target-temperature distribution.

Collision frequency v controls coupling strength

More collisions → faster thermalization
Fewer collisions → dynamics are less disturbed

Random collision with heat bath



Strengths

- Samples the canonical ensemble
- Easy to understand and tune

Limitations

- Random collisions disturb real-time dynamics
- Can distort diffusion and transport properties

Good for canonical sampling, but the random collisions can strongly alter physical dynamics.

3. Berendsen Thermostat

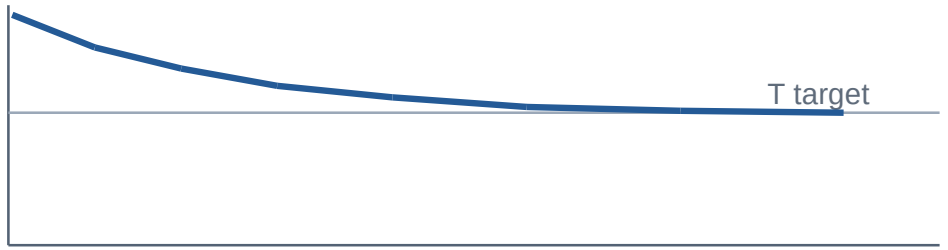
Core idea

Weakly couple the simulated system to a heat bath. Velocities are rescaled smoothly so temperature relaxes toward the target instead of jumping there instantly.

$$dT/dt = (T_0 - T) / \tau T$$

τT = coupling time
small τT → strong/faster coupling

Smooth relaxation toward target temperature



Strengths

- Very stable
- Excellent for early equilibration

Limitations

- Suppresses natural temperature fluctuations
- Does not generate the correct canonical ensemble

Excellent equilibration tool; avoid treating it as rigorous canonical production sampling.

4. Langevin Thermostat

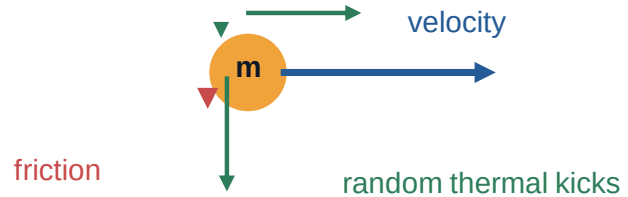
Core idea

Add two heat-bath effects to Newtonian motion: a friction term that removes kinetic energy and a random force that adds thermal energy back.

$$m\,dv/dt = F - \gamma m v + R(t)$$

γ = friction strength
R(t) = random thermal force

Friction + random kicks



Strengths

- Correct canonical sampling when configured properly
- Stable and robust for small systems

Limitations

- Friction can damp real dynamics
- Too-large γ can overdamp motion

This is the family used in our project: Middle Langevin controlled temperature during the accepted simulations.

5. Nosé–Hoover Thermostat

Core idea

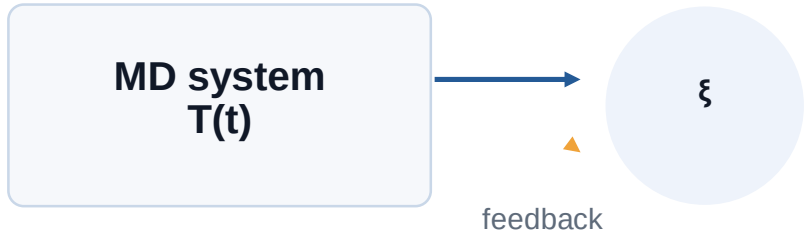
Add an extra dynamical variable that acts like a deterministic friction coefficient. The thermostat variable responds continuously to whether the system is too hot or too cold.

$$\dot{p}_i = F_i - \xi\, p_i$$

ξ grows if T is too high; decreases if T is too low

Nosé–Hoover chains add multiple thermostat variables to improve ergodicity.

Deterministic feedback loop



Strengths

- Deterministic; often preserves dynamics better than random collisions
- Can sample canonical ensemble when ergodic

Limitations

- Can be non-ergodic for small/simple systems
- More parameters and dynamics to tune

A powerful deterministic thermostat, especially with chains, but more subtle than Langevin or Berendsen.

1. Berendsen Barostat

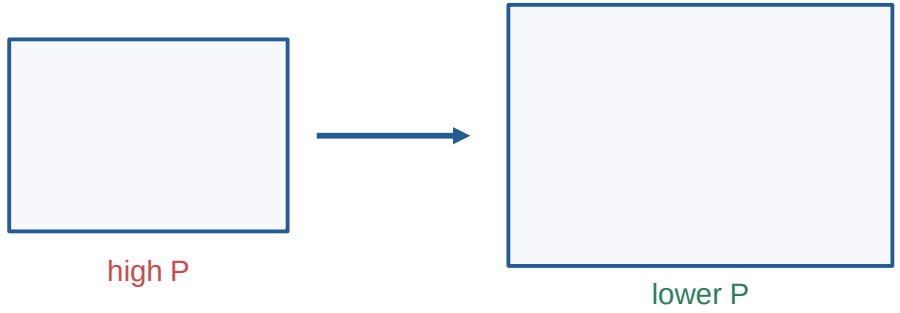
Core idea

Weakly couple the simulation box to a pressure bath. The box volume is smoothly rescaled so the pressure relaxes toward the target.

If pressure is too high → expand the box
If pressure is too low → compress the box

τ_P controls how quickly the volume responds

Smooth box-volume relaxation



Strengths

- Very stable and simple
- Useful for density/volume equilibration

Limitations

- Suppresses correct NPT volume fluctuations
- Not appropriate for rigorous production ensemble sampling

We used Berendsen only as an initial practical pressure-relaxation tool, not as our final production barostat.

2. Parrinello–Rahman Barostat

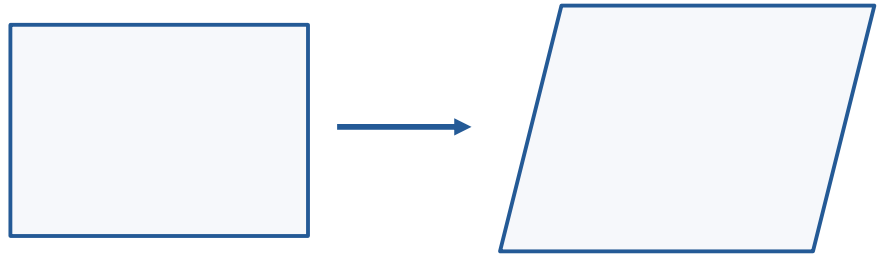
Core idea

Treat the simulation cell itself as a dynamical object. The box vectors can change in length and, when allowed, in shape in response to the pressure/stress tensor.

Useful when anisotropic shape changes matter — especially solids, crystals, interfaces, or phase transformations.

Cell matrix h becomes a dynamical variable

Volume and shape can respond



cell shape changes

Strengths

- Allows physically meaningful box fluctuations and anisotropy
- Widely used for NPT production

Limitations

- Can oscillate if coupling is poorly chosen
- More complex; not always ideal for tiny systems

Unlike Berendsen, Parrinello–Rahman is designed to let the box fluctuate dynamically rather than simply relax it.

3. Bussi / Stochastic Cell-Rescaling Barostat

Core idea

Rescale the box volume using a controlled relaxation toward the target pressure plus a stochastic term that restores the correct NPT volume fluctuations.

volume change = pressure-relaxation term + random fluctuation term

The stochastic part is not “noise for no reason” — it is what helps reproduce the correct ensemble fluctuations.

Fluctuating around the target volume



Strengths

- Designed for correct NPT fluctuations
- Stable and practical for production

Limitations

- Still requires sensible coupling parameters
- Stochastic box dynamics are not intended to reproduce a literal physical piston

This was our final barostat: Bussi pressure coupling for the accepted NPT equilibration and production stages.

4. Monte Carlo Barostat

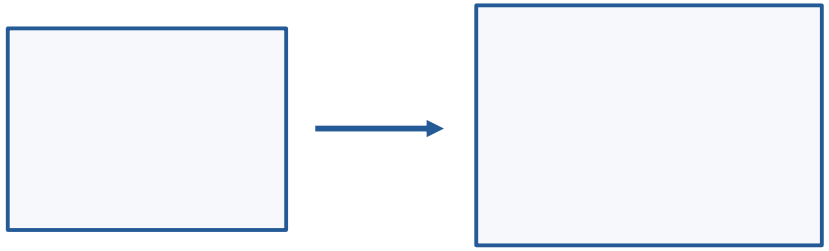
Core idea

Instead of continuously moving the box, periodically propose a trial volume change. Recalculate the energetic/statistical cost, then accept or reject the new box according to the NPT probability.

propose V′ → evaluate → accept / reject

Accepted move → continue from V′
Rejected move → keep the old box

Discrete trial move



Strengths

- Samples NPT ensemble without fictitious box dynamics
- Conceptually clean acceptance rule

Limitations

- Each volume proposal can require expensive energy work
- Not a continuous physical pressure trajectory

Useful when correct equilibrium sampling matters more than having continuous barostat dynamics.

Which Methods Did We Use in Our Water Project?

Temperature control

Middle Langevin

- Stable canonical temperature control
- Handles small-system temperature fluctuations well
- Used throughout the accepted heating / equilibration / production stages

Why? Robust temperature control without the fluctuation-suppression problem of Berendsen.

Pressure / volume control

Berendsen — early relaxation

Bussi — final NPT

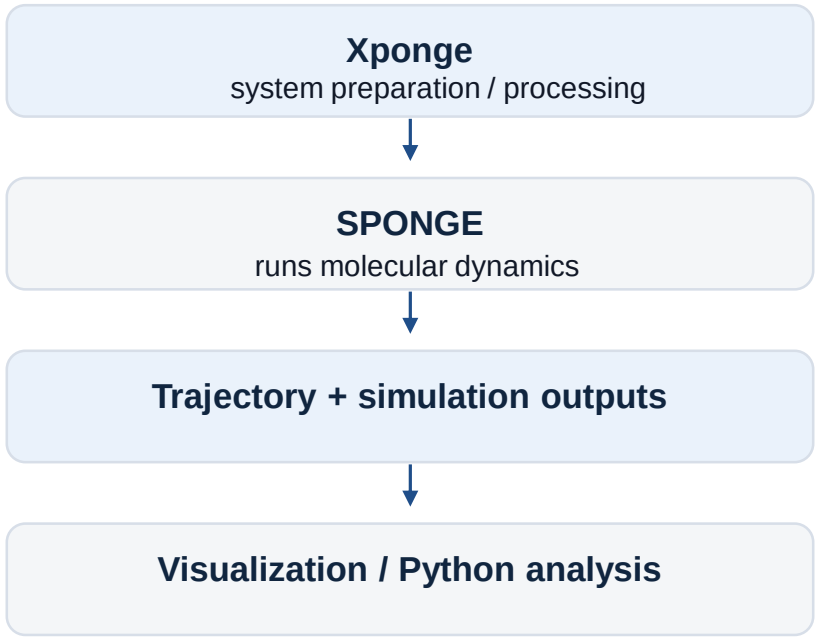
- Berendsen: gently move density/volume toward a reasonable region
- Bussi: retain appropriate NPT fluctuations for final equilibration and production

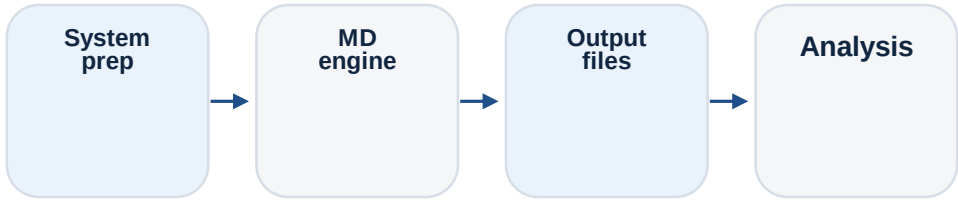
Protocol logic: relax first → sample properly later.

Thermostats and barostats are not interchangeable: they control different physical variables and can generate different ensembles.

Practical MD Tools

The learning phase also moved from theory to real simulation tools.





Xponge → SPONGE → VMD / MDAnalysis

VMD → visualization
MDAnalysis / Python → trajectory analysis

The practical-learning section connected Xponge → SPONGE → visualization/analysis tools.

Where AI Fits Into Molecular Dynamics

Traditional MD:

coordinates → analytical force field → forces → integrator

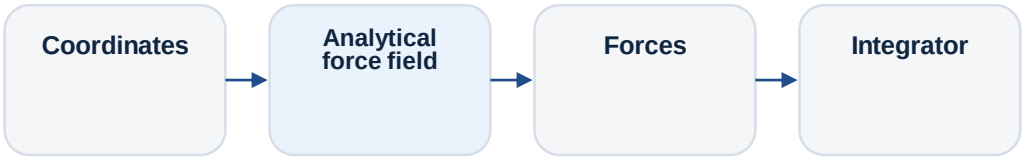
AI-based MD:

coordinates → learned potential → forces → integrator

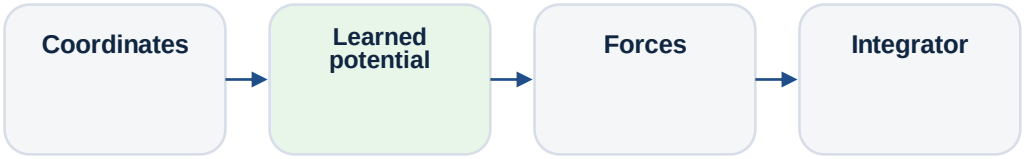
AI may replace the energy/force model — not the whole MD engine.

Periodic boundaries, integration, ensembles, thermostats, stability and analysis are still required.

Traditional loop



AI-assisted loop



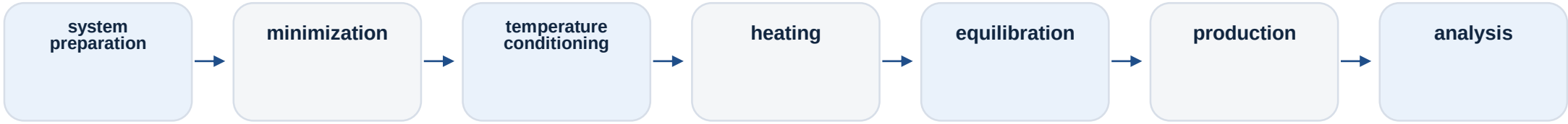
This distinction was a key conclusion of the learning phase.

Moving From Theory to My First MD Experiment

After learning the MD workflow, the next objective was to implement it end-to-end.

Project objective

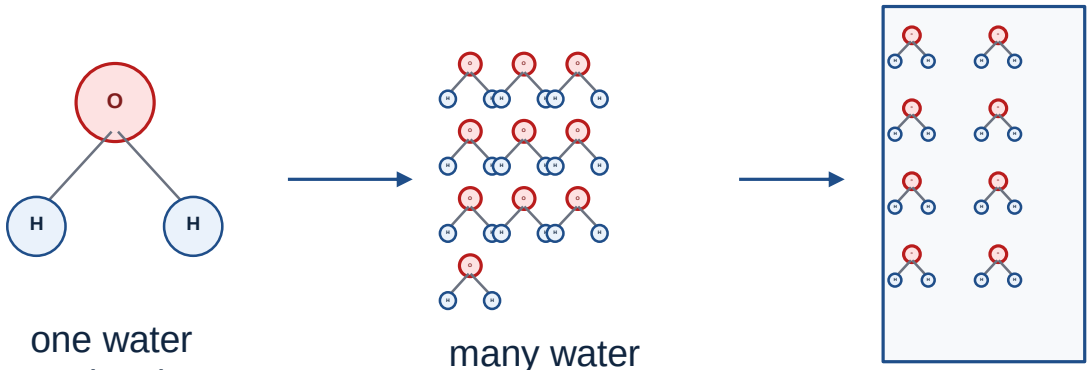
Build a periodic liquid-water system and complete:



Why Start With Water?

Water was a useful first complete molecular simulation because:

- it is physically important;
- molecular geometry is well defined;
- established classical water models exist;
- liquid structure can be analyzed using RDFs;
- it contains the main ingredients of realistic MD: electrostatics, van der Waals interactions, constraints, PBC and pressure control.



Choosing the TIP3P Water Model

We selected TIP3P, a three-site rigid water model.

One molecule contains:

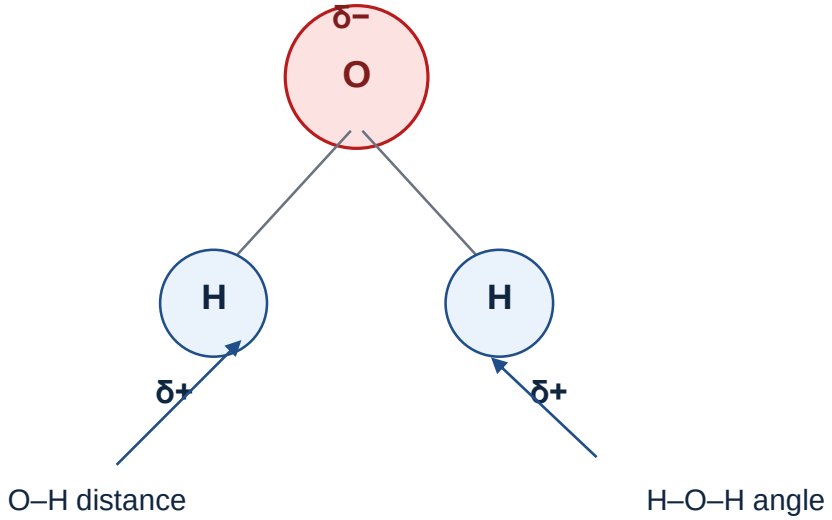
- 1 oxygen
- 2 hydrogen atoms

Final system:

862 water molecules

862 × 3 = 2586 atoms

The final input files preserve the O–H–H atom ordering for all 862 molecules.



Building the Simulation Box

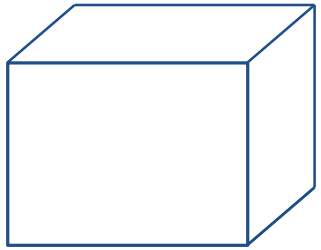
Initial periodic system:

- 862 TIP3P water molecules
- 2586 atoms
- approximately 30 × 30 × 30 Å³
- initial density ≈ 0.9551 g/cm³

Process

Generate molecules
→ place them inside box
→ check atom ordering/topology
→ check overlaps
→ create periodic system

PLACEHOLDER
Insert actual VMD-style screenshot of the 30 Å water box here

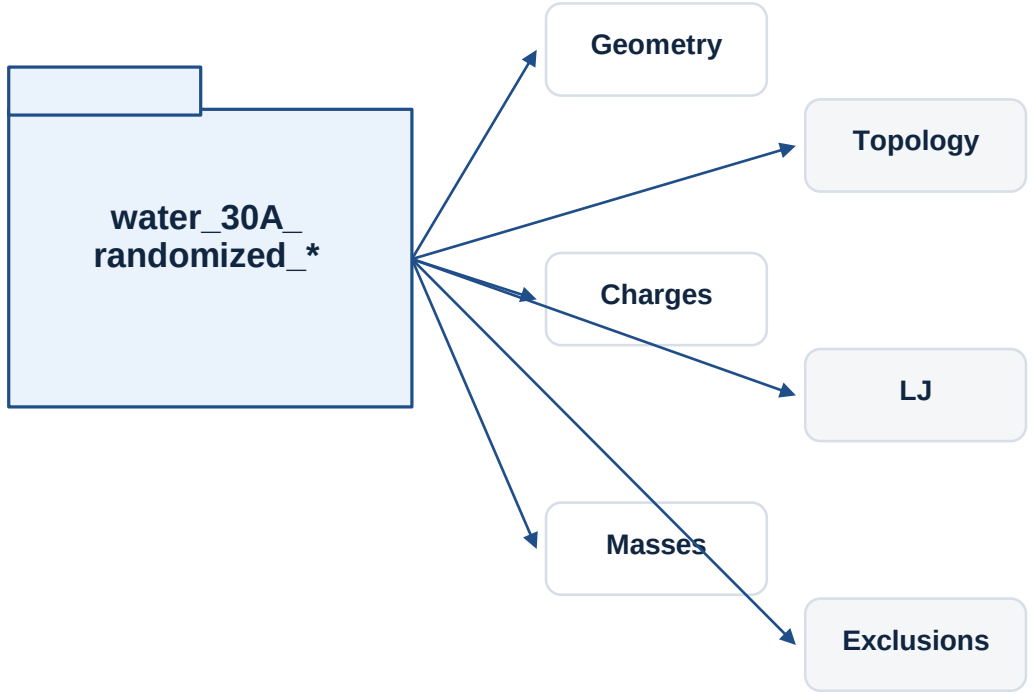


~30 Å

From the Water Box to SPONGE Input Files

The molecular system was converted into SPONGE input files describing:

- coordinates
- atom names/types
- masses
- charges
- Lennard-Jones parameters
- bonds / constraints
- exclusions
- residue information



In backup, we can show the complete input-file table.

Final Simulation Parameters

This should be one of the cleanest technical slides.

Parameter	Final choice
Water model	TIP3P
Molecules	862
Atoms	2586
Initial box	~30 × 30 × 30 Å³
Target temperature	300 K
Target pressure	1 bar
Main timestep	**1 fs**

Parameter	Final choice
Non-bonded cutoff	**8 Å**
Long-range electrostatics	**Particle Mesh Ewald**
Water constraints	**SETTLE**
Thermostat	**Middle Langevin**
Initial pressure relaxation	Berendsen
Final NPT equilibration / production	Bussi pressure coupling

SPONGE was requested with a SHAKE constraint flag, but the runtime initialized all 862 rigid waters through SETTLE, so SETTLE is the correct constraint method to report.

Simulation Environment

Platform

Windows + PowerShell

Environment

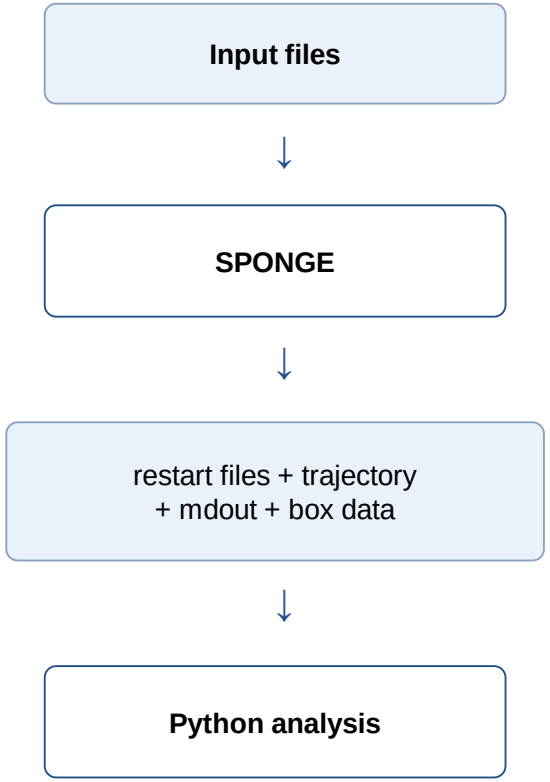
sponge-cpu

MD engine

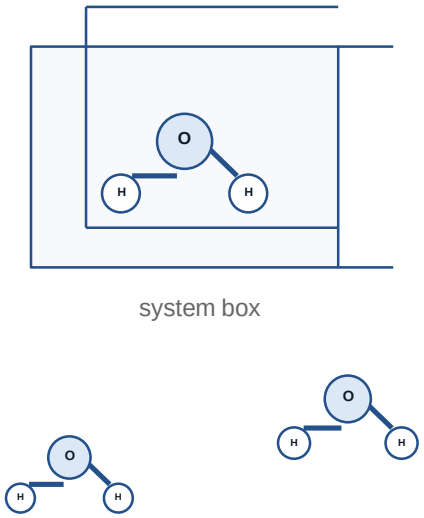
SPONGE

Supporting package

Xponge



Simple architecture of the practical SPONGE workflow

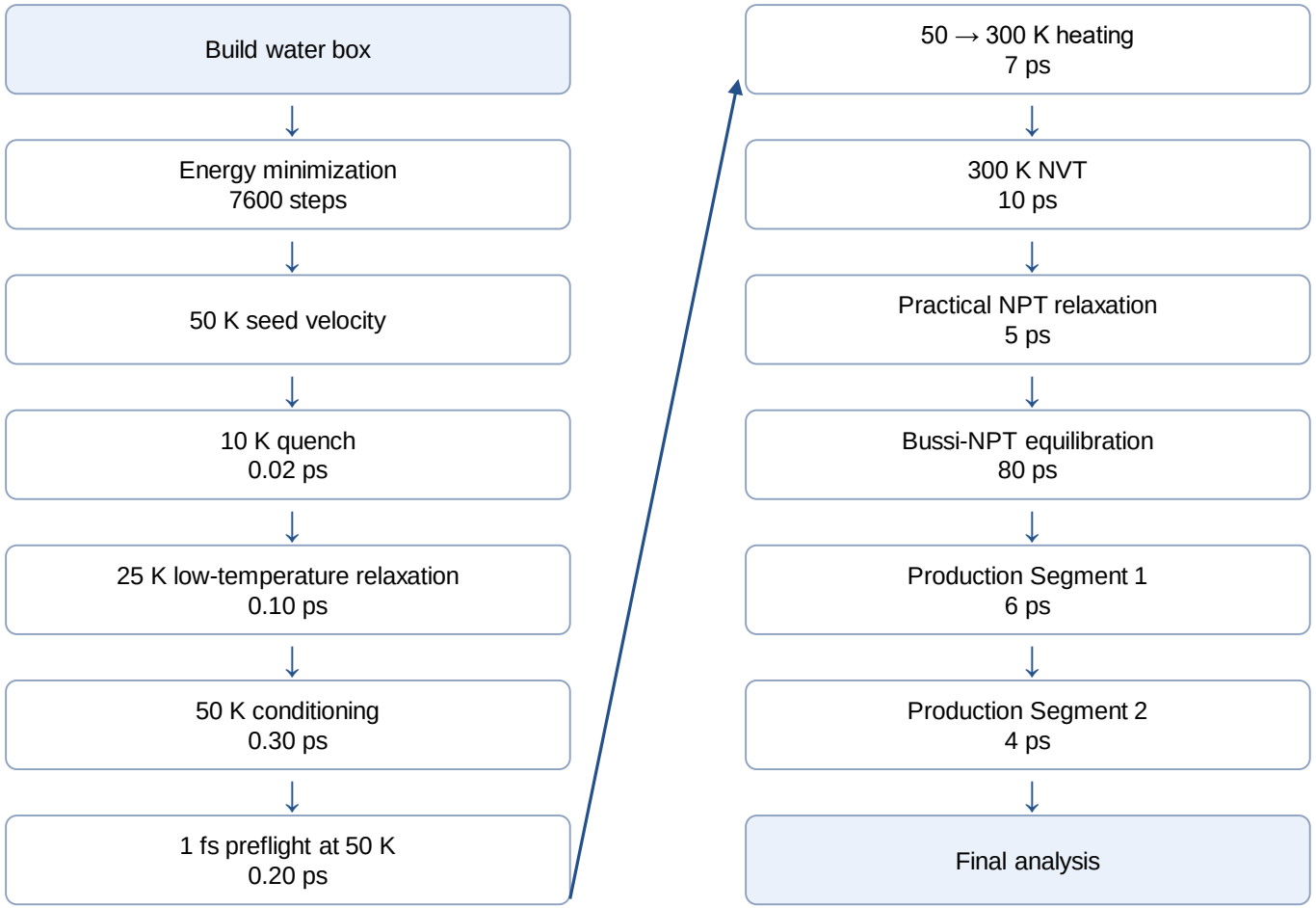


system box

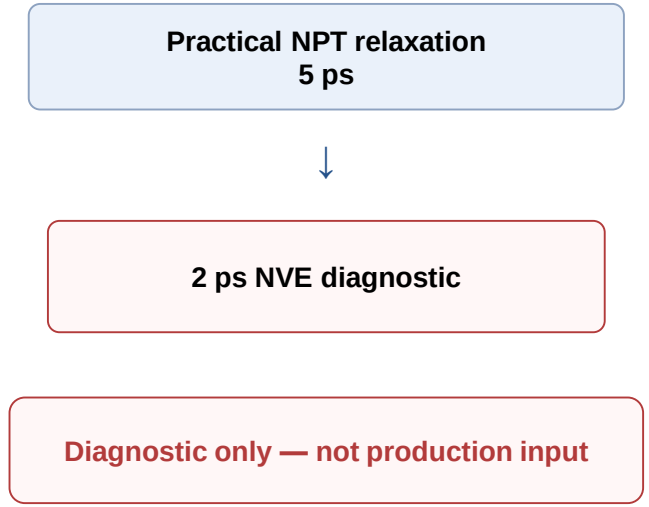
Do not show a big terminal screenshot unless you have a clean one.

The Complete Accepted Simulation Workflow

Main production chain



Side branch from the 5 ps practical NPT state



Stage 1: Energy Minimization

Why?

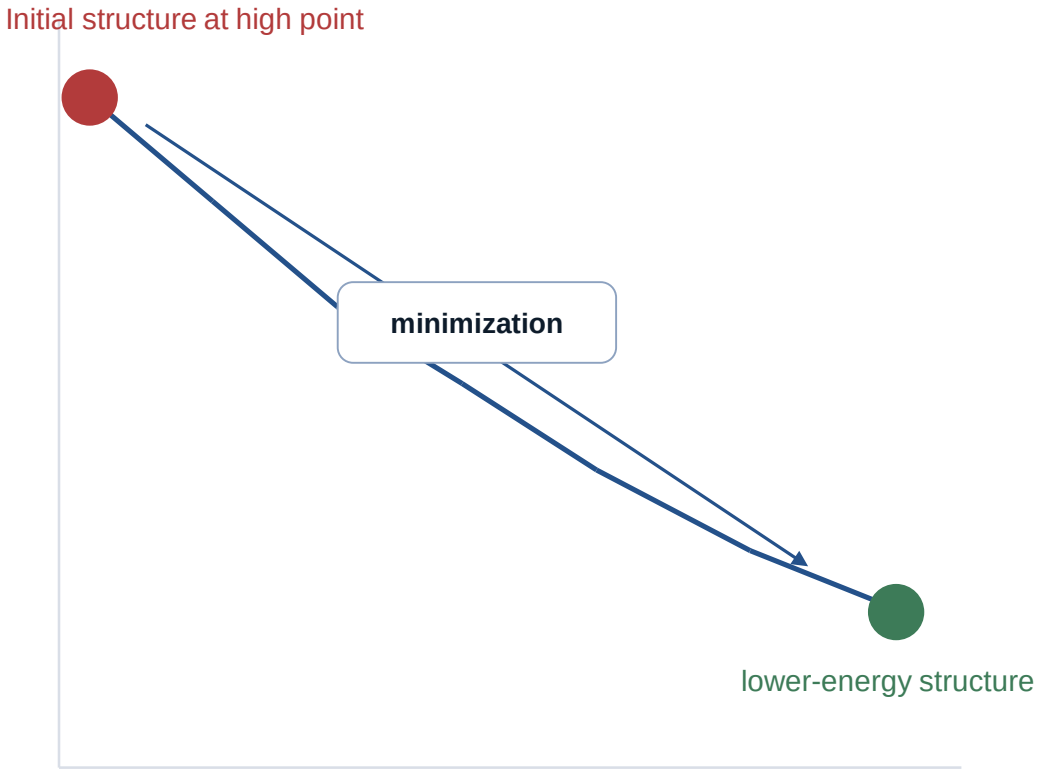
The constructed box may contain unfavorable local geometry or excessive forces.

What we did

Accepted minimization: 7600 cumulative steps

Goal

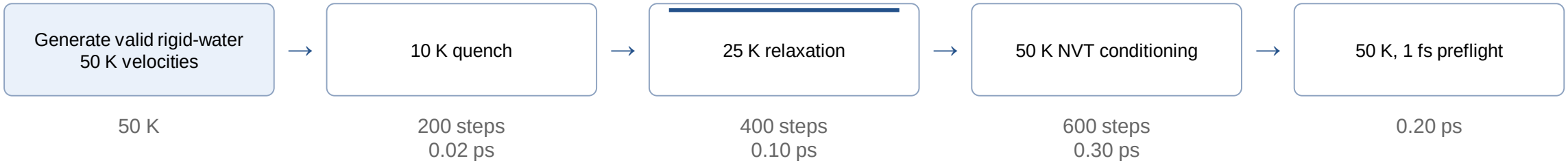
reduce problematic forces and moves atoms toward a safer, lower-energy configuration..



Potential energy landscape

Before minimization → After minimization

Stage 2: Initial Velocities and Low-Temperature Conditioning



Temperature staircase: 50 K seed → 10 K → 25 K → 50 K → 50 K preflight

The 25 K state is not claimed to be a fully equilibrated 25 K system.
It was accepted as a safe low-temperature state for continuing to 50 K.

Stage 3: Controlled Heating to 300 K

Heating too rapidly can destabilize a newly prepared system by excessive velocities, unstable forces, and structural problems.

So temperature was increased gradually.

Actual heating protocol

50 K starting state

→ 100 K — 1 ps

→ 150 K — 1 ps

→ 200 K — 1 ps

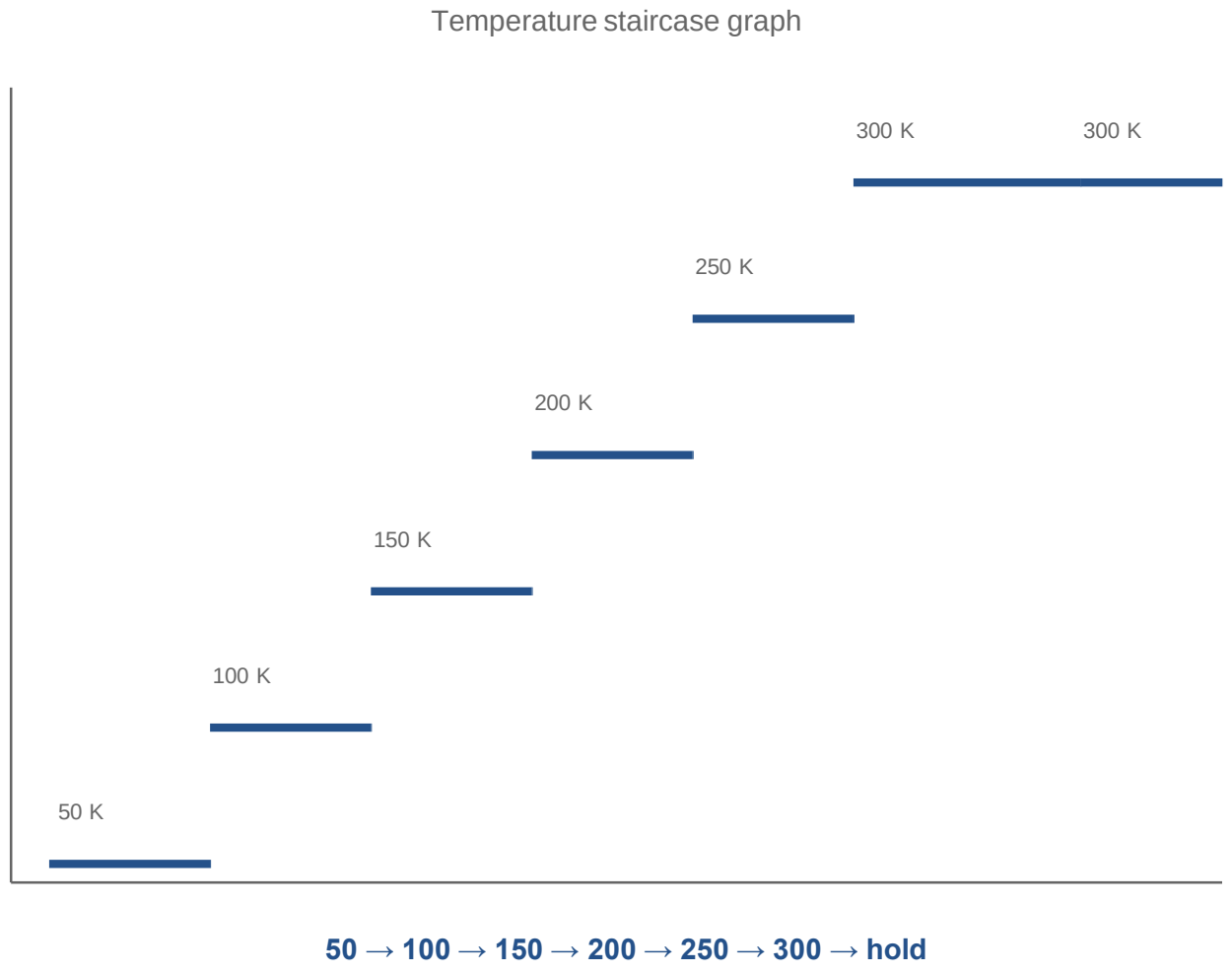
→ 250 K — 1 ps

→ 300 K — 1 ps

→ 300 K hold — 2 ps

Total

7 ps



Stage 4: 300 K NVT Stabilization

After heating, the system was held at:

300 K

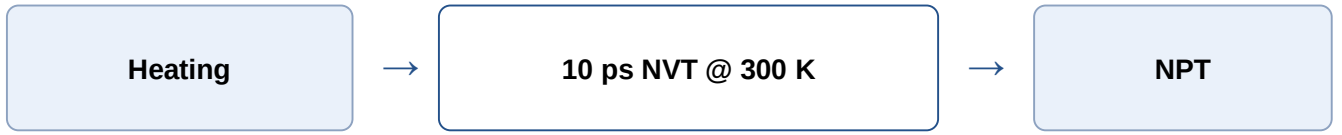
under the NVT ensemble

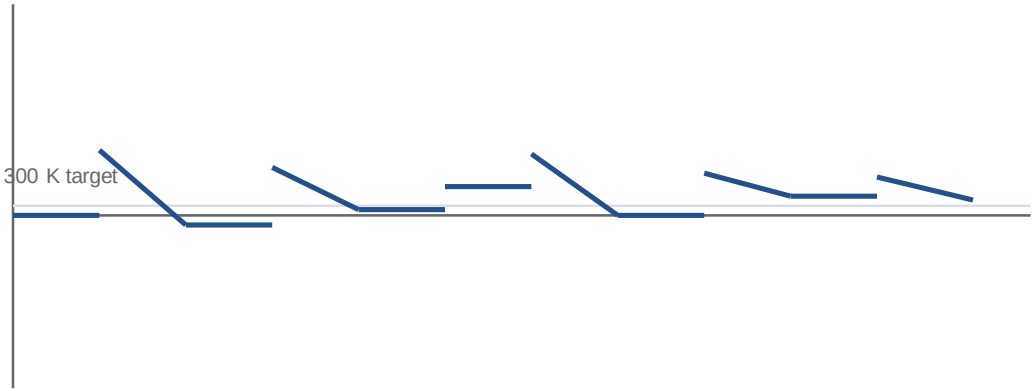
for:

10 ps

Purpose

- allow temperature to stabilize;
- relax the heated configuration;
- prepare a stable input for pressure relaxation.





schematic temperature fluctuations around 300 K

Stage 5: Practical NPT Density Relaxation

The next goal was to allow the box volume and density to respond to pressure.

Conditions

NPT

300 K

1 bar target

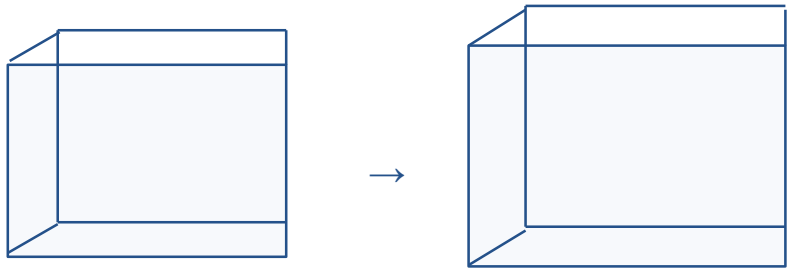
5 ps practical relaxation

Method

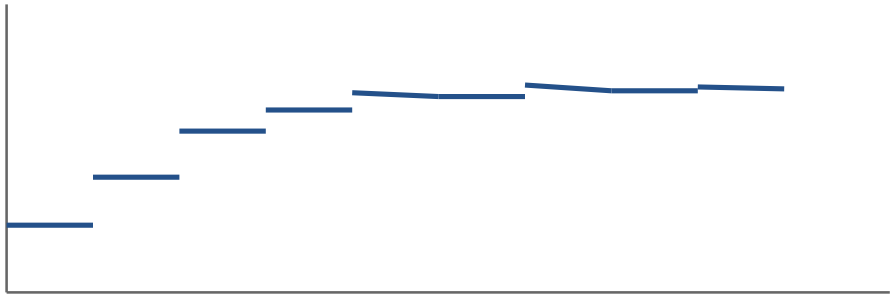
Middle Langevin thermostat

• initial Berendsen pressure relaxation

Berendsen was used as a practical relaxation tool, not as evidence of rigorous production-ensemble sampling.



initial size slightly adjusted size



density moving toward a stable region

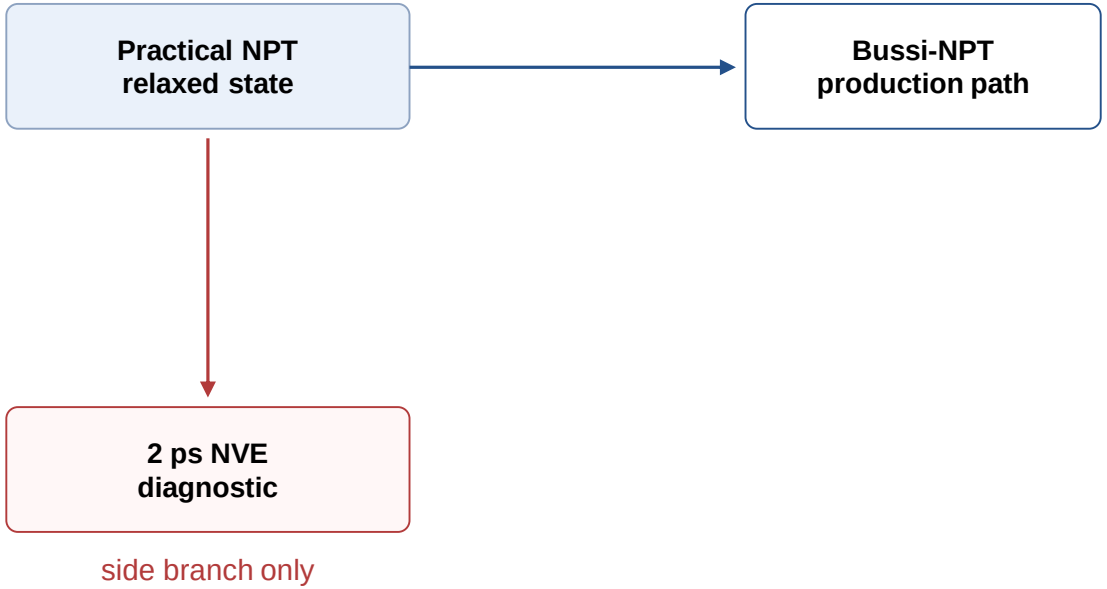
Stage 6: NVE Numerical-Stability Diagnostic

From the relaxed NPT state, we also ran a separate:

2 ps NVE diagnostic

Why?

To check for obvious short-time numerical or structural instability without active temperature/pressure control.



NVE was diagnostic only.
It was not used as the input to production.

Stage 7: 80 ps Bussi-NPT Equilibration

For the final pre-production equilibration:

Ensemble: NPT

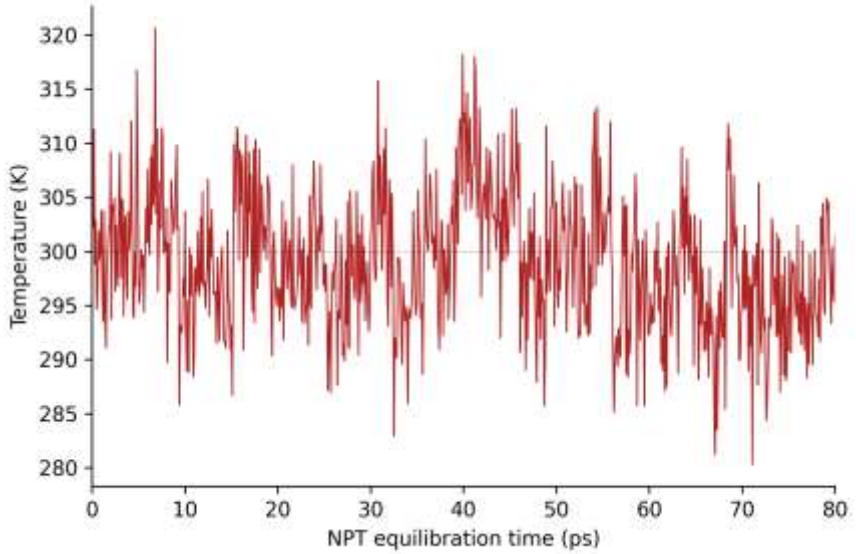
Temperature: 300 K

Pressure target: 1 bar

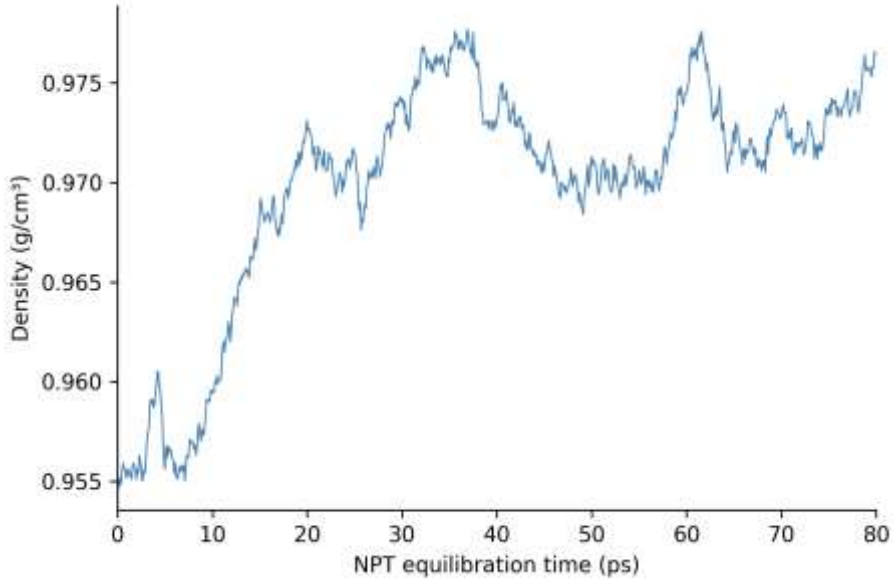
Duration: 80 ps accepted

Why this stage mattered

It was much longer than the earlier relaxation (5ps) and allowed the volume/density behavior to be assessed before production.



npt_equilibration_80ps_temperature



npt_equilibration_80ps_density

Stage 8: Final Production Simulation

Production had to be completed in two validated segments.

Segment 1

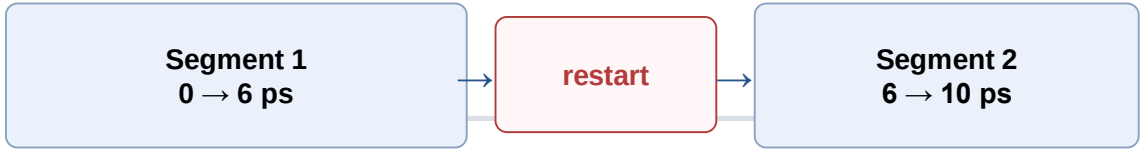
6 ps accepted

The later unsynchronized tail was excluded after an unexpected laptop/system restart.

Segment 2

Restarted from the synchronized Segment-1 endpoint

4 ps



→ **final production trajectory**

6 ps + 4 ps = 10 ps

We combine them for analysis

Challenges and How the Protocol Evolved

This is valuable because an internship is also about solving problems. Do not make it a diary of every failed pass.

Runtime limits

Original longer trajectories were computationally impractical on the available CPU.

Decision: use shorter scientifically controlled stages.

Low-temperature initialization

The early state required additional conditioning.

Decision: validated 50 K seed + staged conditioning.

Interrupted production

Laptop/system restart interrupted Segment 1.

Decision: identify synchronized restart → continue with Segment 2.

RDF normalization

Initial RDF calculation was not conventionally normalized.

Decision: audit analysis → recompute proper dimensionless intermolecular g(r).

The final RDF calculation deliberately excludes same-molecule O–H and H–H pairs and correctly approaches g(r) ≈ 1 at large separation.

Final Production Analysis Window

Final production: 10 ps

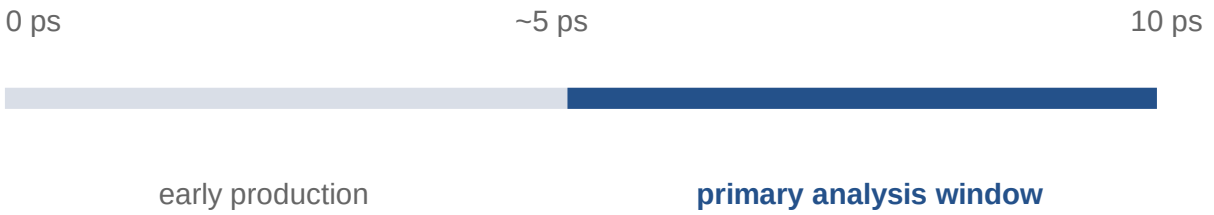
Primary thermodynamic window:

107.72−112.62 ps

approximately the final 4.9 ps

Thermodynamic records: 50

Accepted structural trajectory frames: 14



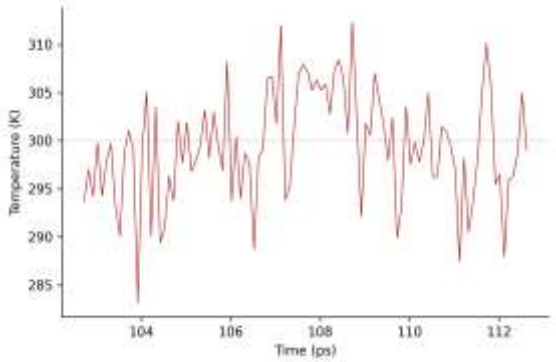
Highlight the last ~5 ps.

Results are suitable for a short-timescale practical MD study, not for strong long-timescale statistical conclusions.

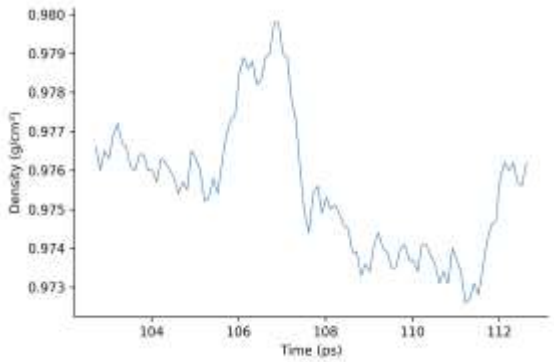
Final Thermodynamic Results

Quantity	Mean ± SD
Temperature	**300.386 ± 5.682 K**
Pressure	**9.37 ± 483.60 bar**
Density	**0.974238 ± 0.000979 g/cm³**
Potential energy	**−8203.52 ± 34.10 kcal/mol**
Box volume	**26468.73 ± 32.50 Å³**

These values are the frozen final production-level results.



temperature



density

Main interpretation

Temperature remained near the 300 K target.

Density was stable around 0.974 g/cm³ during the short primary window.

Pressure, Potential Energy and Volume

Pressure

9.37±483.60 bar

Pressure fluctuated strongly.

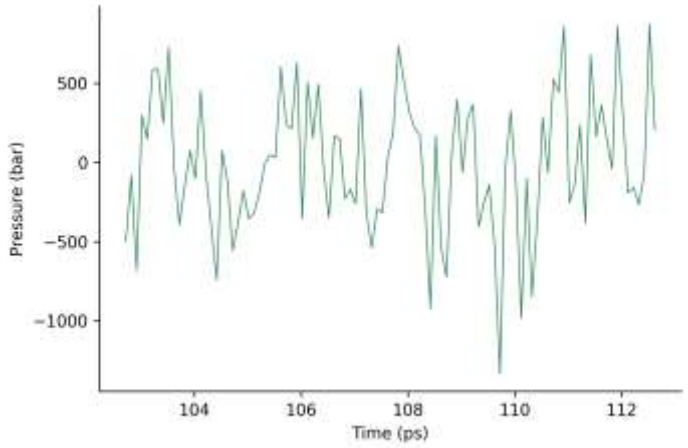
Therefore:

the mean pressure should not be interpreted as proof of complete pressure equilibration.

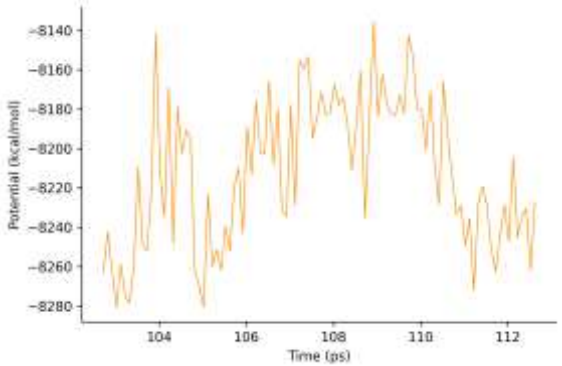
Other behavior

Potential energy remained finite and stable over the short analysis window.

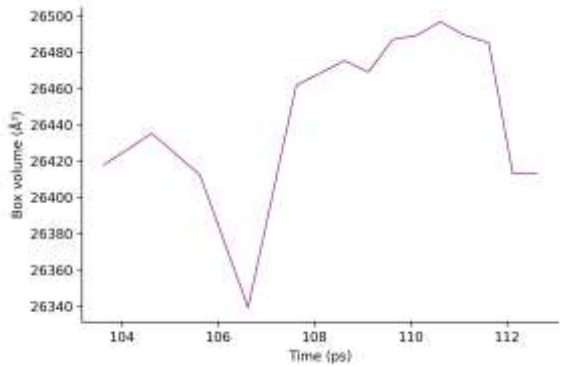
Volume showed relatively small fluctuations.



pressure



potential

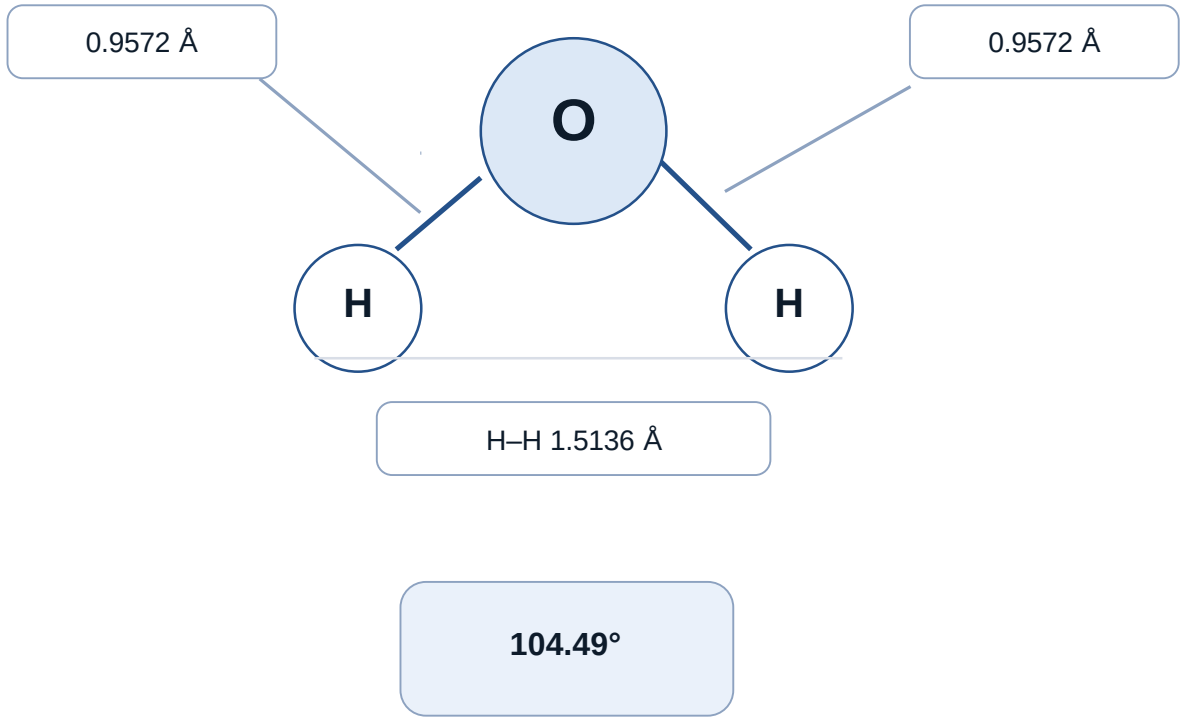


box_volume

Did the TIP3P Molecular Geometry Stay Correct?

Property	Mean
O–H distance	**0.957197 Å**
H–H distance	**1.513595 Å**
H–O–H angle	**104.4906°**

The measured deviations were extremely small.

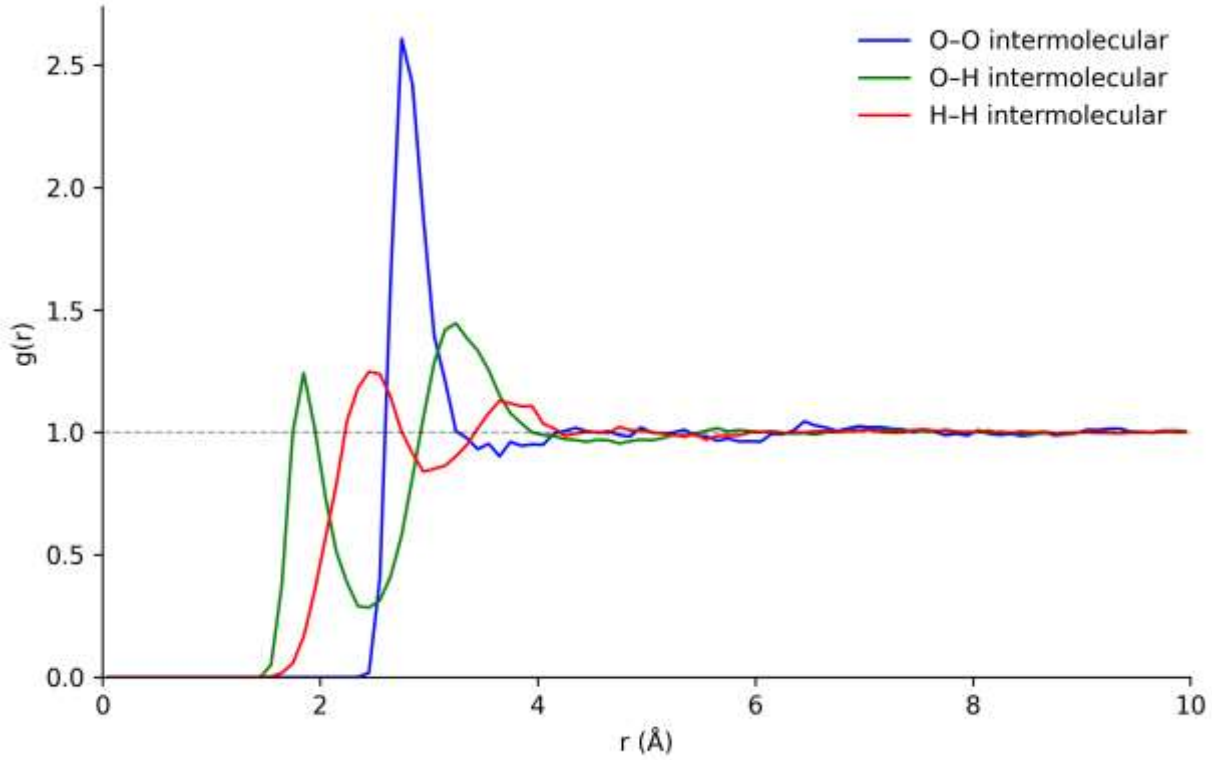


Interpretation

The rigid-water geometry remained essentially fixed.

This is consistent with SETTLE correctly maintaining TIP3P geometry.

Liquid Structure: Radial Distribution Functions



rdf_intermolecular_normalized

Pair	First local peak	g(r)
O–O	**2.75 Å**	**2.607**
O–H intermolecular	**1.85 Å**	**1.242**
H–H intermolecular	**2.45 Å**	**1.247**

At large r:
g(r) → 1

The O–O peak near 2.75 Å shows a preferred nearest-neighbor oxygen separation and local liquid-water organization.

14 frames only → qualitative/illustrative structural interpretation.

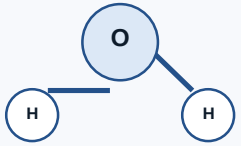
What the Results Tell Us

1. Temperature



The system stayed centered near the 300 K target.

2. Molecular structure



TIP3P geometry remained stable and physically reasonable.

3. Liquid organization



The RDF shows short-range structural organization, especially the O–O first shell around 2.75 Å.

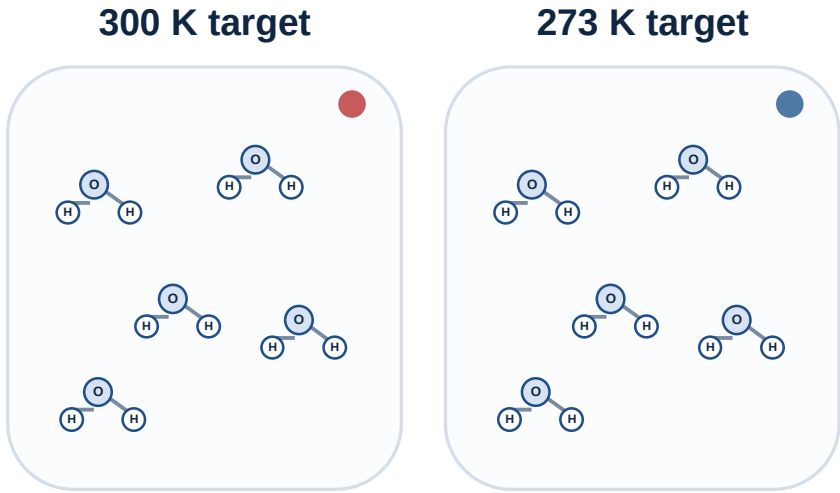
The complete workflow produced a physically reasonable short-timescale liquid-water simulation.

Experiment 2: How Temperature Changes Liquid-Water Structure

Matched TIP3P Molecular Dynamics at 300 K and 273 K

The first experiment established a complete, validated TIP3P liquid-water MD workflow at 300 K.

After reviewing the results, the next objective was to go beyond basic thermodynamics and RDFs and ask how lowering temperature changes the liquid-water interaction network and local structure.



same molecules • same box style • only temperature differs

The Main Question

How does lowering the target temperature from 300 K to 273 K change the thermodynamic behavior, hydrogen-bond network and local molecular structure of liquid TIP3P water?

Thermodynamics

Temperature
Density
Potential energy
Volume



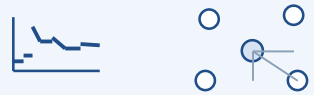
Hydrogen-bond network

Population
Distance
Angle



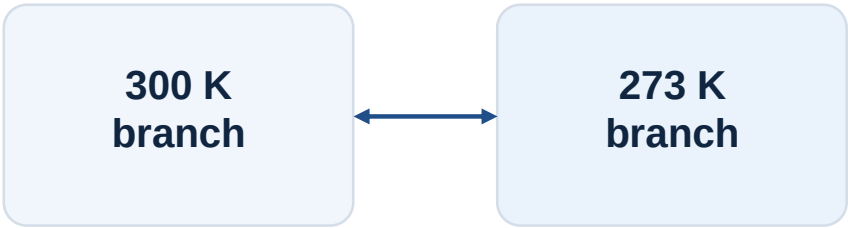
Local structure

OO / OH / HH RDF
Tetrahedral order
Coordination



What Changed — and What Stayed the Same?

Kept identical	Changed
TIP3P water	Target temperature
862 waters / 2586 atoms	300 K → 273 K
Same force field	Thermostat seed
NPT / 1 bar	Required restart provenance
1 fs timestep	
PME	
8 Å cutoff	
SETTLE rigid water	
Middle Langevin	
Bussi barostat	
5 ps production	
0.05 ps structural sampling	

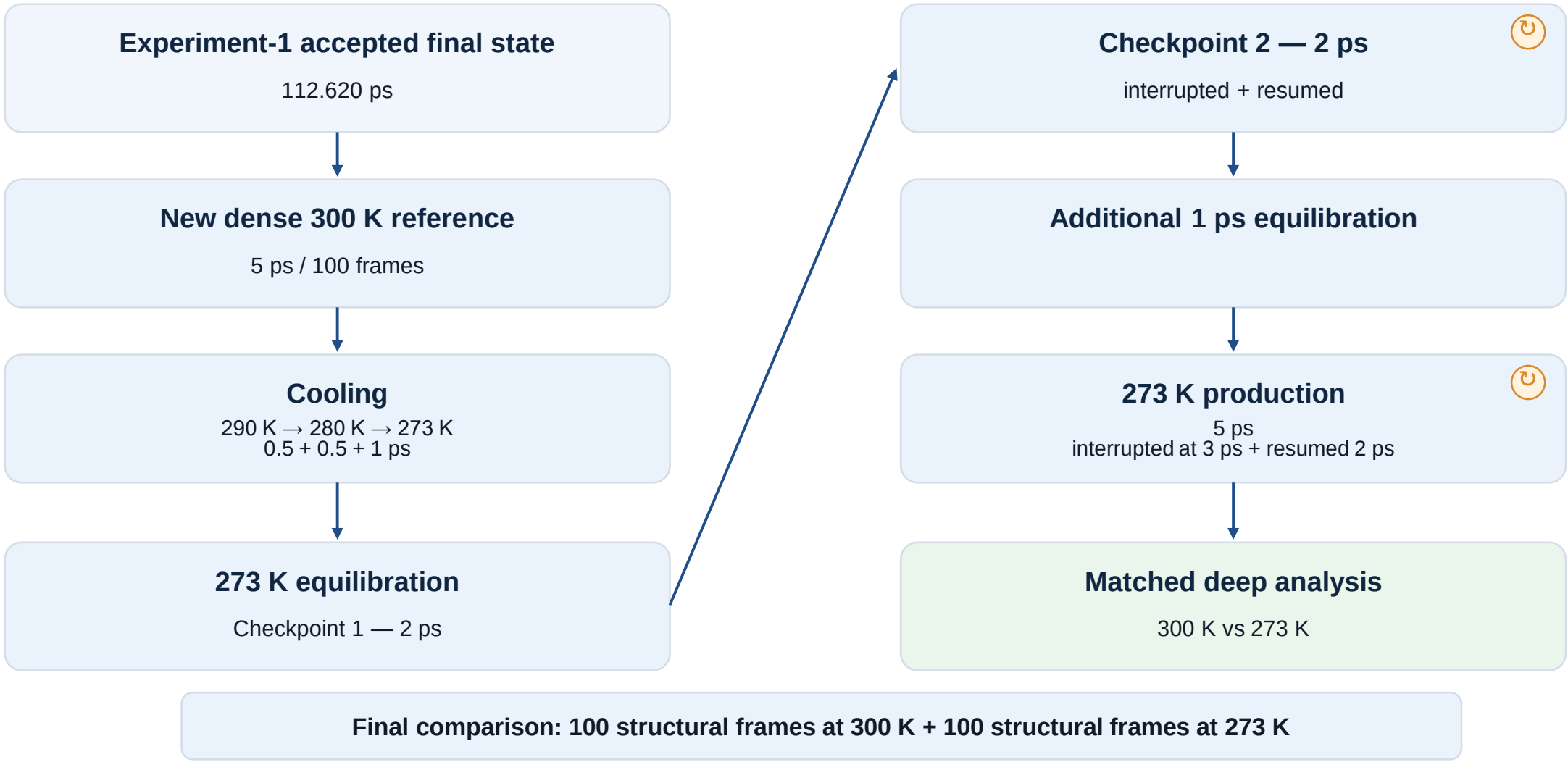


**same system +
same methodology**

Only the experimental temperature branch and unavoidable provenance details changed.

The goal was a matched comparison, so temperature was changed while the physical and numerical setup remained as similar as possible.

The Complete Experiment 2 Workflow



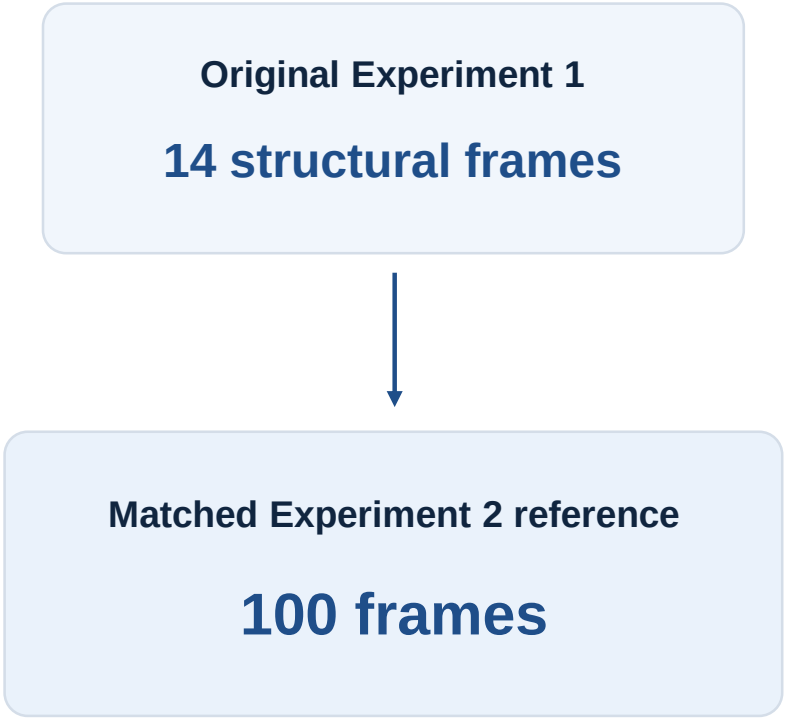
Stage 1: Building a Matched 300 K Reference

The original Experiment-1 structural analysis contained only 14 accepted structural frames.

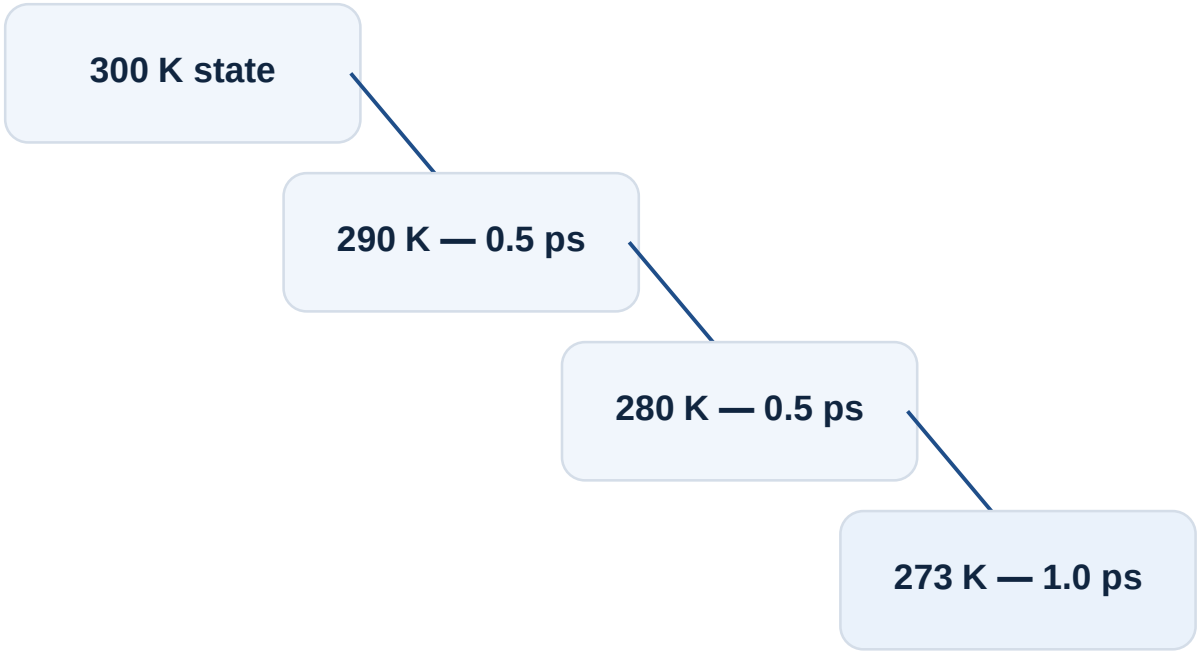
That was sufficient for an illustrative RDF, but not ideal for a direct dense comparison with a new 273 K trajectory.

So we generated a new matched reference:

Parameter	Value
Target temperature	300 K
Ensemble	NPT
Duration	5 ps
Steps	5000
Timestep	1 fs
Sampling interval	0.05 ps
Trajectory frames	100
Box records	100
Thermo records	100



Stage 2: Controlled Cooling



Stage	Observed mean T
290 K target	293.54 K
280 K target	289.38 K
273 K target	284.51 K

Cooling protocol

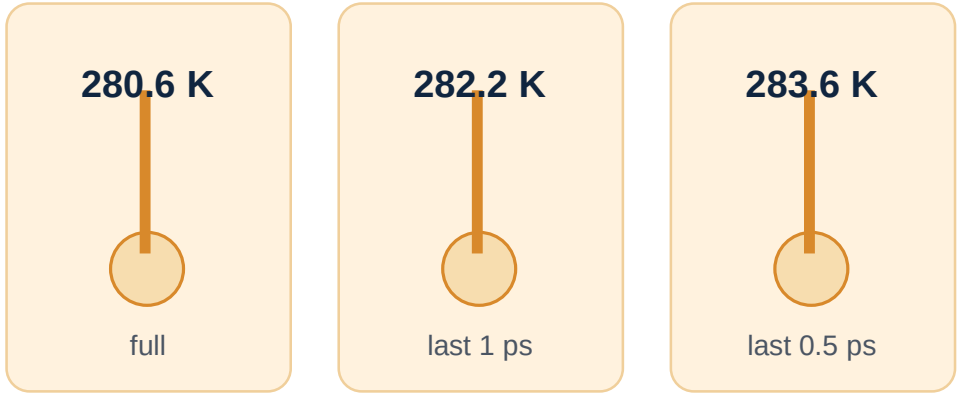
300 K → 290 K → 280 K → 273 K

Cooling was completed successfully, but the final state was not yet thermally equilibrated around 273 K.

Stage 3: 273 K Equilibration — Checkpoint 1

NPT · 273 K target · 1 bar · 2 ps · same physical settings

Window	Temperature
Full 2 ps	280.61 ± 4.58 K
Last 1 ps	282.21 K
Last 0.5 ps	283.57 K



Computationally valid — thermally not accepted

The temperature remained well above the target, so production was not started.

Checkpoint 2: Continue Equilibration — Then Another Interruption

Planned: 2 ps additional equilibration at 273 K



synchronized restart available at 123.120 ps

Recovery

Instead of repeating all 2 ps, we resumed only the missing 0.5 ps.

After recovery

Last 1 ps: 278.93 ± 2.93 K
Last 0.5 ps: 278.45 ± 2.91 K

Temperature had become stable, but around 278–279 K, still above the 273 K target.

Before Running More MD, We Audited the Thermal Behavior

Target temperature

All accepted segments used 273 K

Thermo parsing

Correct temperature column

Degrees of freedom

Correct rigid-water DOF treatment

Thermostat configuration

No configuration error found

Density was nearly flat, while potential energy was still evolving, suggesting that the system had not fully finished relaxing.

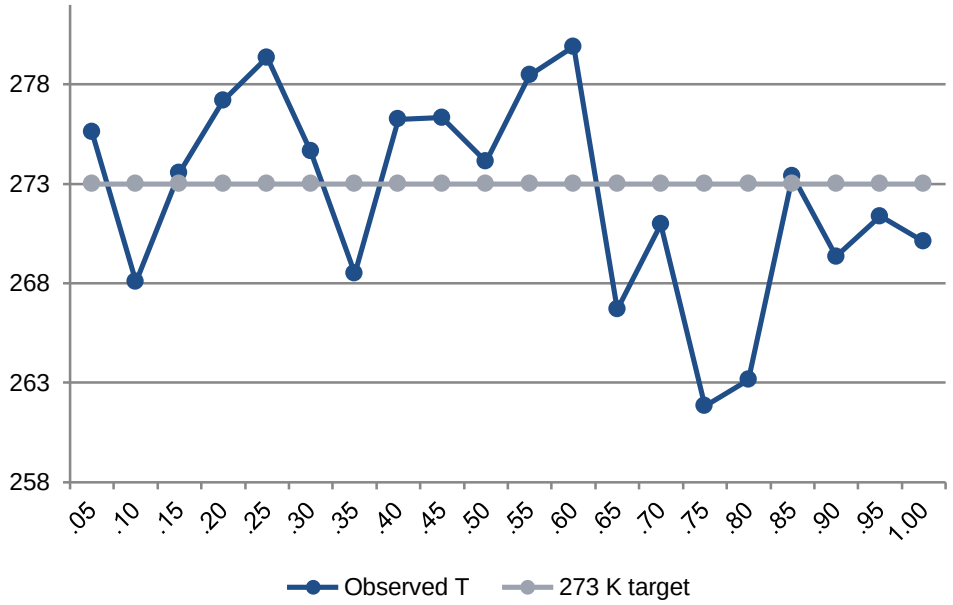
We did not change the thermostat or force the temperature downward artificially.

NO CONFIGURATION PROBLEM
→ extend equilibration by only 1 ps

The audit separated a genuine relaxation problem from a configuration or parsing failure.

The Final 1 ps Extension Reached the Operational Target

Quantity	Result
Full 1 ps temperature	272.44 ± 5.11 K
Last 0.5 ps	270.52 ± 5.84 K
Density	essentially flat
Potential-energy drift	leveled off



T3_273K_EQUILIBRATION_OPERATIONALLY_ACCEPTED

Accepted for beginning matched production. Not claimed as rigorous long-timescale equilibrium.

Stage 4: Matched 273 K Production

Parameter	Final choice
Target T	273 K
Target P	1 bar
Ensemble	NPT
Duration	5 ps
Steps	5000
Timestep	1 fs

Parameter	Final choice
PME cutoff	8 Å
Constraint	SETTLE
Thermostat	Middle Langevin
Barostat	Bussi
Structural interval	0.05 ps
Intended frames	100

This matched the new 300 K reference except for temperature, seed and required restart provenance.

The 5 ps Production Was Also Interrupted



At the interruption:

- synchronized restart existed at 127.620 ps
- structural frame 60 was already safely written
- box record 60 was safely written
- the step-3000 thermo record was still buffered and was lost

Recovery

Resume from the synchronized step-3000 restart for only the remaining 2 ps.

Final Production Dataset

Trajectory

60 + 40 = 100 frames

Box

100 records

Thermodynamics

59 + 40 = 99 records

One genuine record at 127.620 ps is absent.

T4_273K_PRODUCTION_SEGMENTED_VALIDATED

Final simulation time: 129.620 ps

The Analysis

RDF



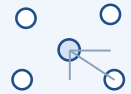
OO
OH
HH
secondary features

Hydrogen bonds



count vs time
distance
angle

Local order



tetrahedral q
coordination

Thermodynamics



temperature
density
potential
volume

Both temperatures were recalculated with the same analysis settings and the same 100-frame structural sampling.

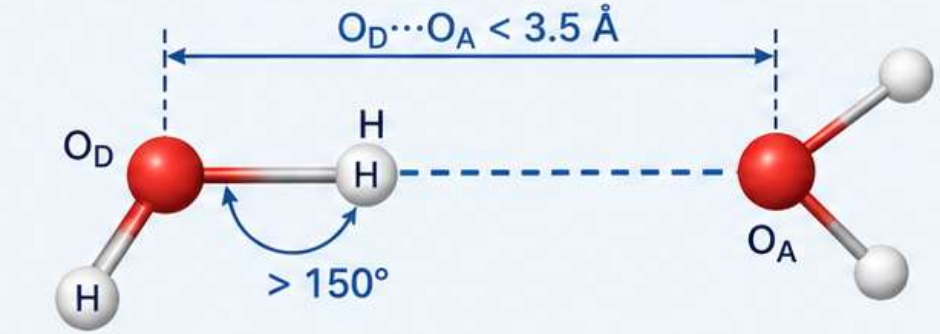
How We Defined Hydrogen Bonds and Tetrahedral Order

Hydrogen bond

$$O_D \cdots O_A < 3.5\ \text{Å}$$

$$\angle\, O_D\text{–}H \cdots O_A > 150°$$

- angle vertex = H
- acceptor belongs to a different water
- minimum-image PBC
- counted as donor-O + donor-H + acceptor-O triplets

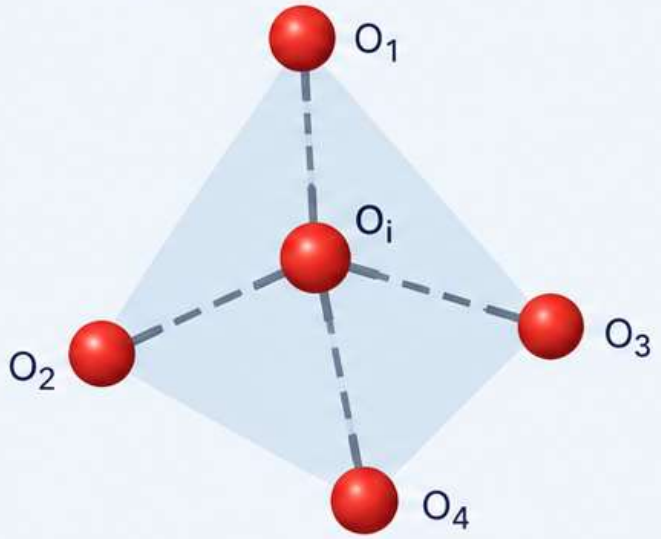


Tetrahedral order

$$q_i = 1 - (3/8) \sum_{j<k}^{4} (\cos \psi_{jik} + 1/3)^2$$

Four nearest oxygen neighbors under PBC.

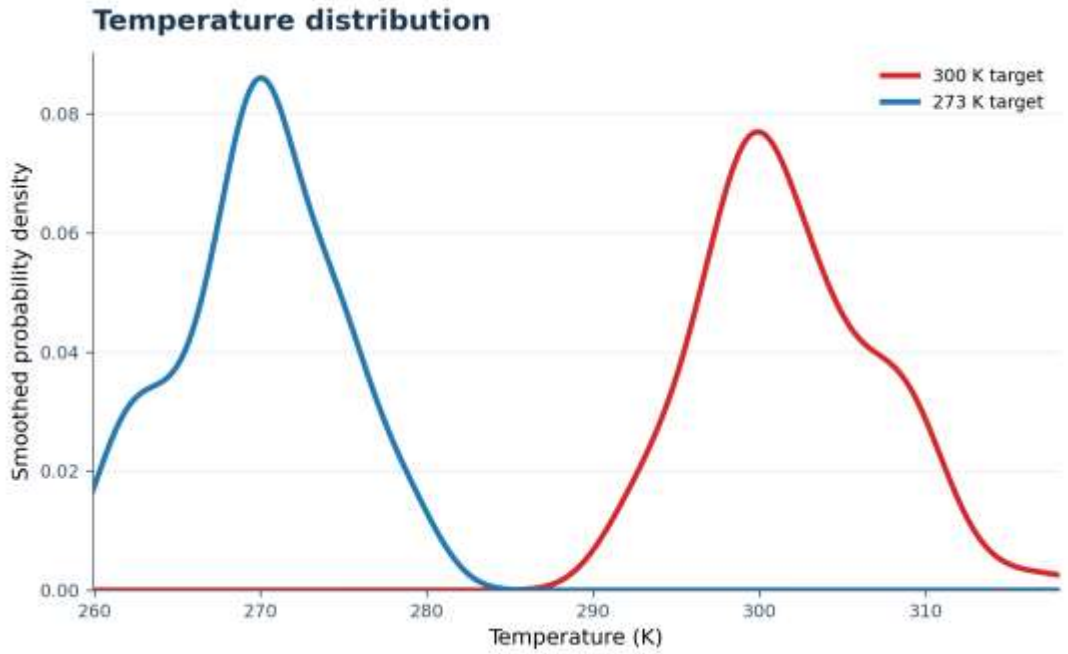
$q \approx 1$ → highly tetrahedral local arrangement.

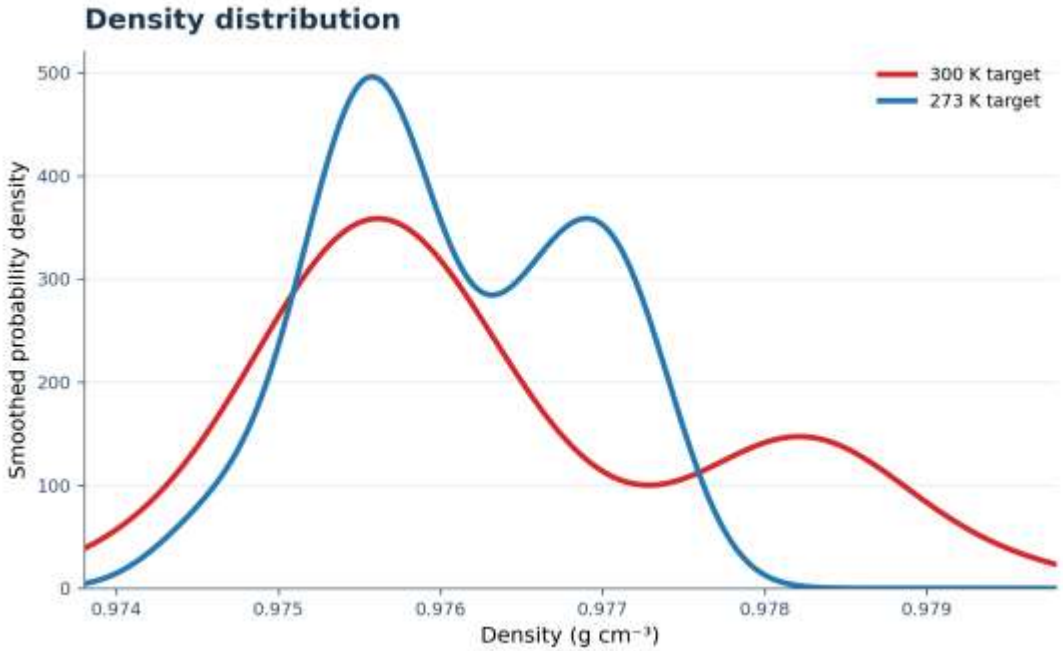


Thermodynamic Comparison: Temperature and Density

	300 K target	273 K target
Sampled T	301.79 ± 5.30 K	269.96 ± 4.76 K
Density	0.976319 ± 0.001366	0.976066 ± 0.000786 g/cm³

The two branches clearly sampled different thermal regimes.
Mean density remained very similar over these short trajectories.



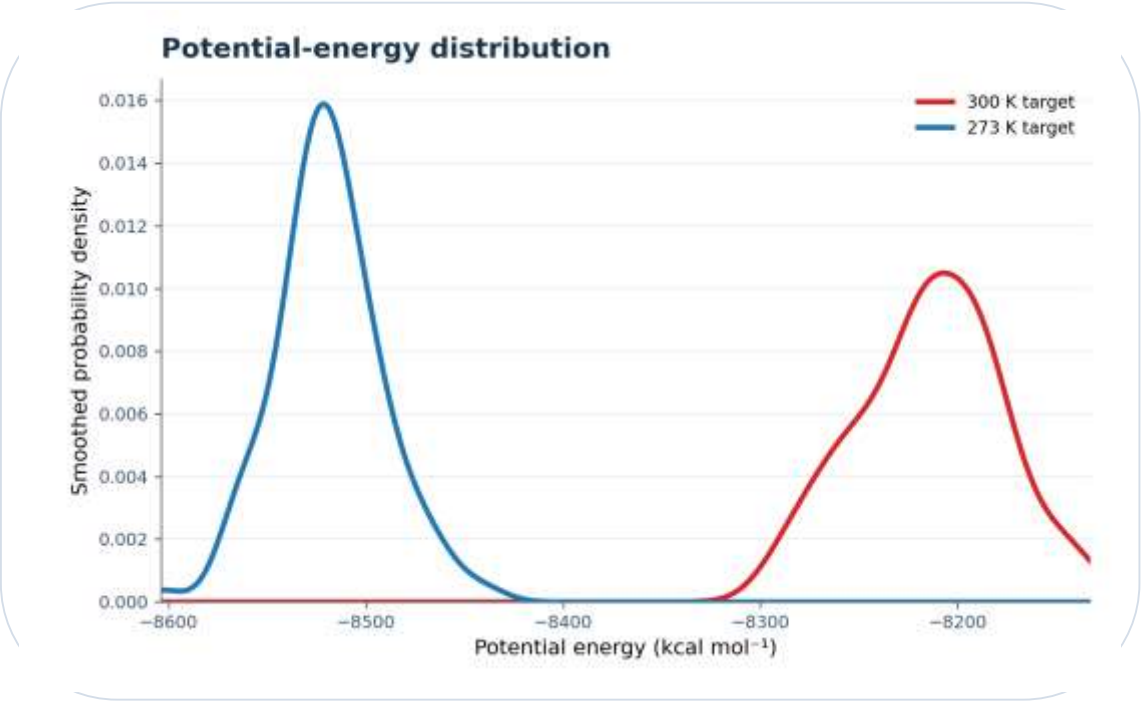


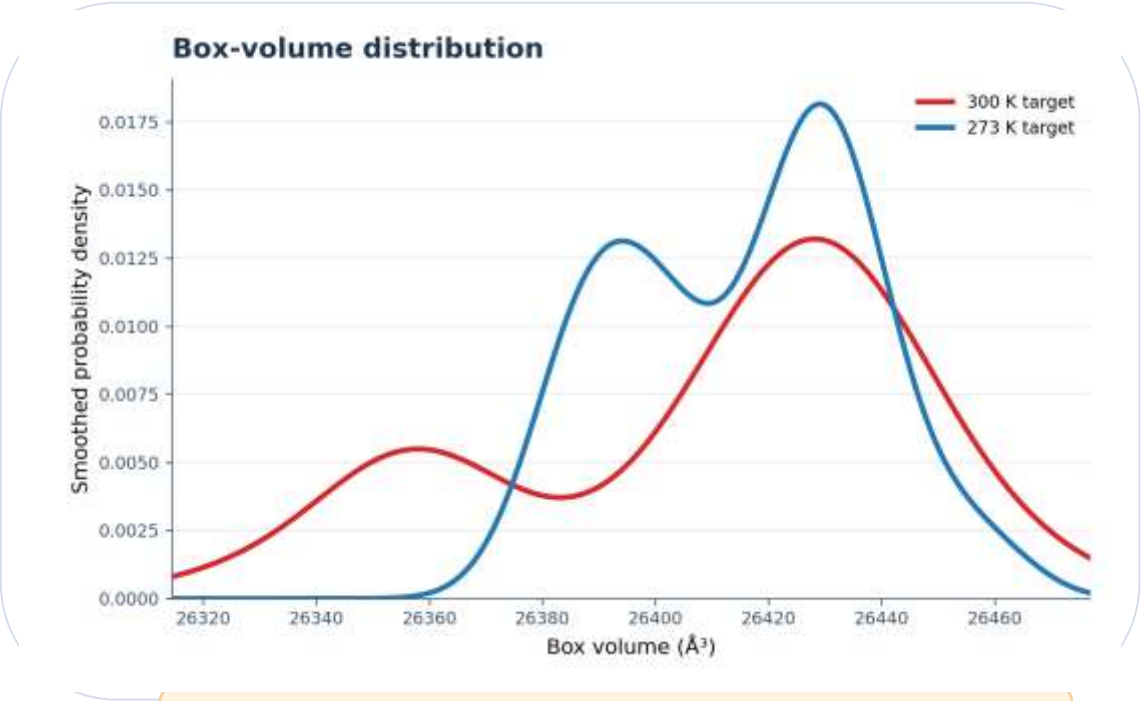
Energy and Box Behavior

Quantity	300 K target	273 K target
Potential energy	−8214.39 ± 36.21	−8518.88 ± 27.19 kcal/mol
Box volume	26409.27 ± 36.92	26415.94 ± 21.32 Å³
Pressure	−75.25 ± 395.94	−592.70 ± 357.78 bar

The cooler branch had a substantially lower potential energy.

Mean volume remained very similar.

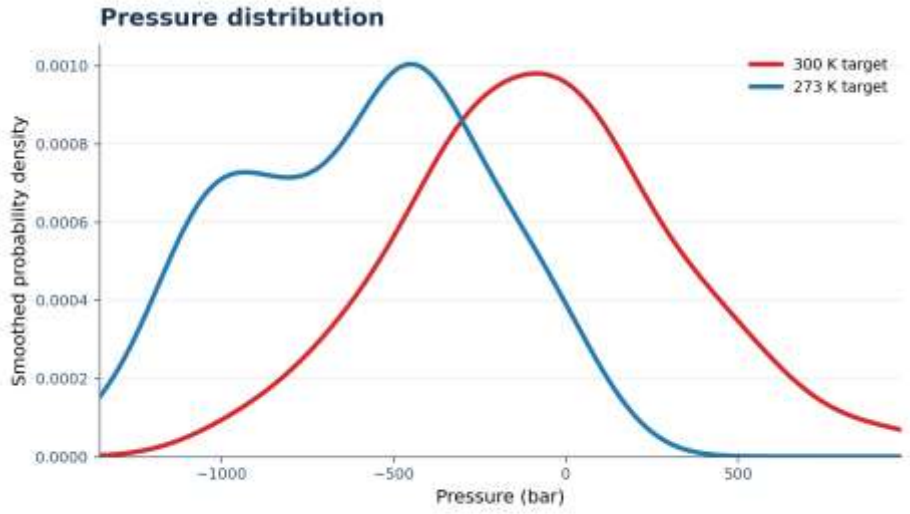




We have exactly the same 862 water molecules in both systems, so if volume hardly changes, density also hardly changes.

Pressure Distribution

Quantity	300 K target	273 K target
Pressure	−75.25 ± 395.94 bar	−592.70 ± 357.78 bar
Interpretation	large fluctuations	large fluctuations



Instantaneous NPT pressure is extremely noisy in small-system MD. Its mean is not strong evidence of equilibration.

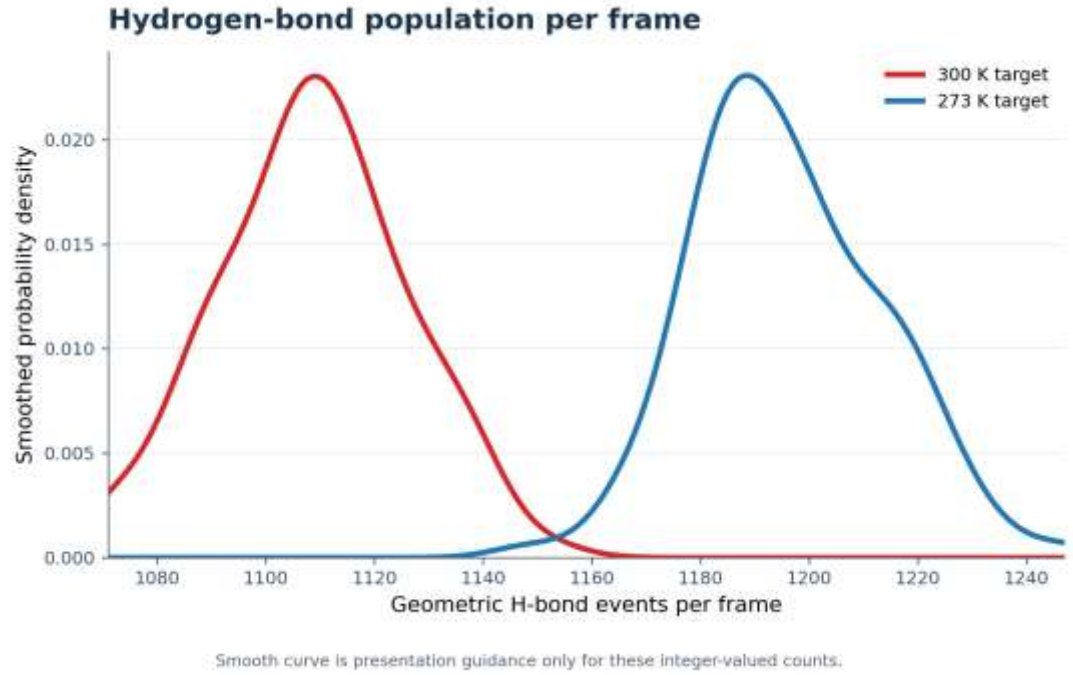
Pressure is shown descriptively only; it was not used as the main acceptance criterion.

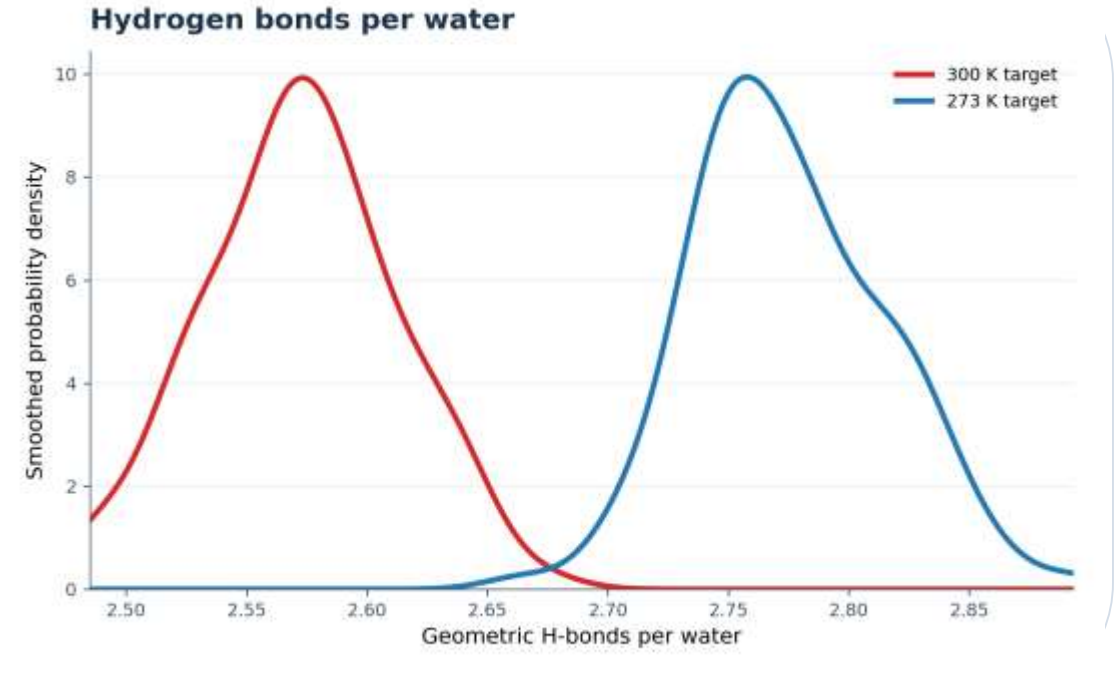
The broad pressure distributions reflect strong small-system fluctuations, so pressure is reported only descriptively.

Cooling Produced More Geometric Hydrogen Bonds

	300 K target	273 K target
H-bond events/frame	1108.70 ± 17.00	1195.86 ± 17.19
H-bonds/water	2.5724 ± 0.0394	2.7746 ± 0.0399

Approximately 7.9% more geometric H-bond events in the cooler-target trajectory.

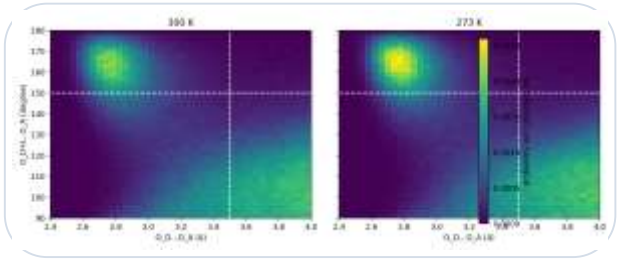




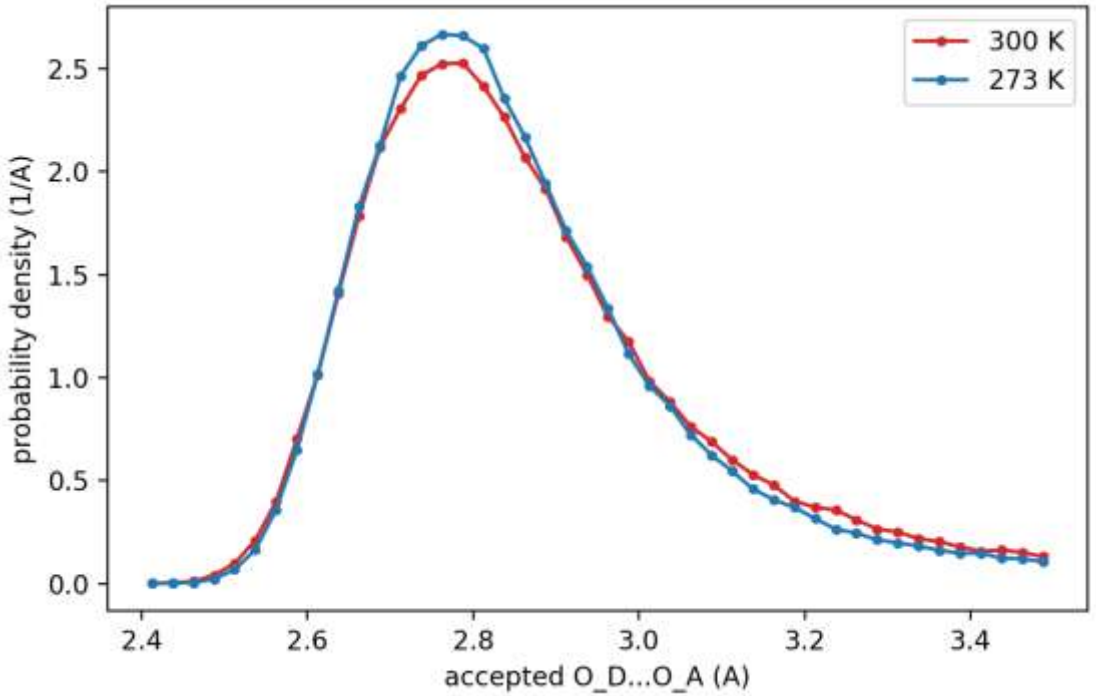
The Cooler Network Was Slightly Shorter and Straighter

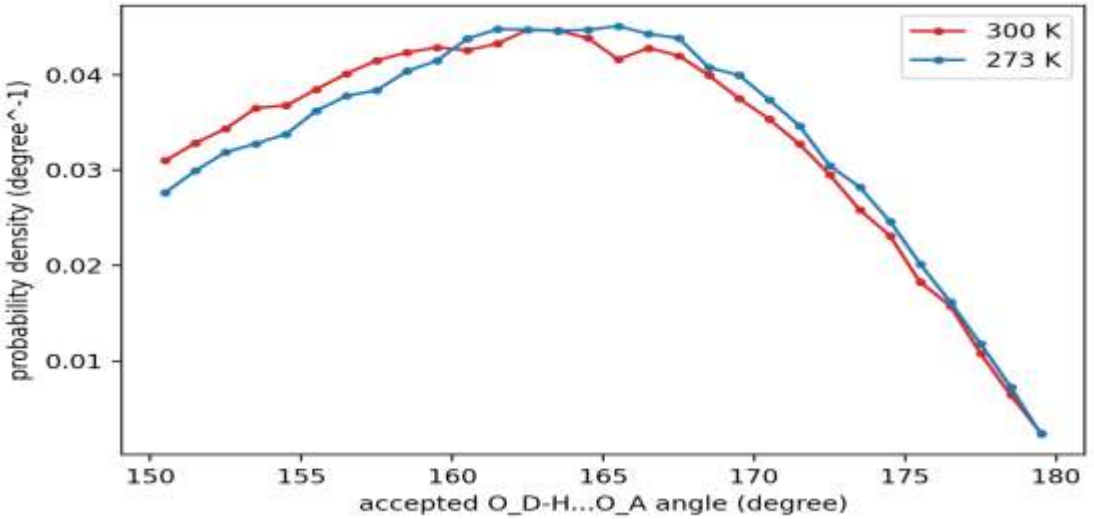
	300 K	273 K
Accepted O···O	2.8614 ± 0.1910 Å	2.8523 ± 0.1809 Å
O–H···O angle	162.99 ± 7.31°	163.39 ± 7.28°

Accepted H-bond geometries shifted slightly toward shorter O···O distances and more linear angles.



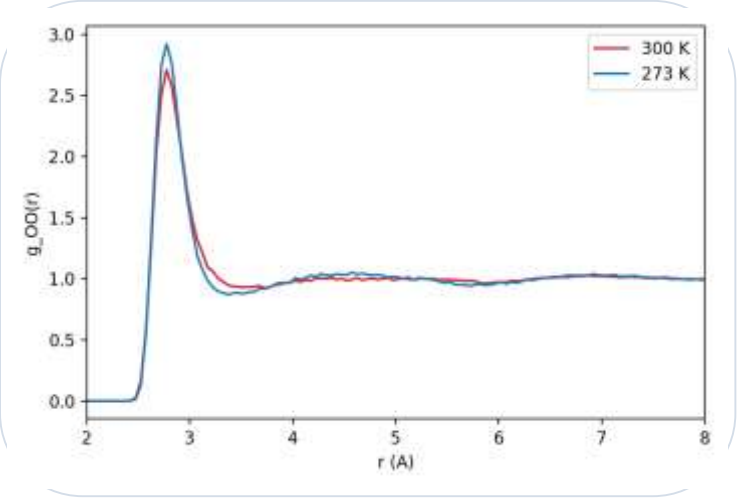
This is consistent with a somewhat more ordered geometric network, not a direct measurement of H-bond energy.





How Local Liquid Structure Changed

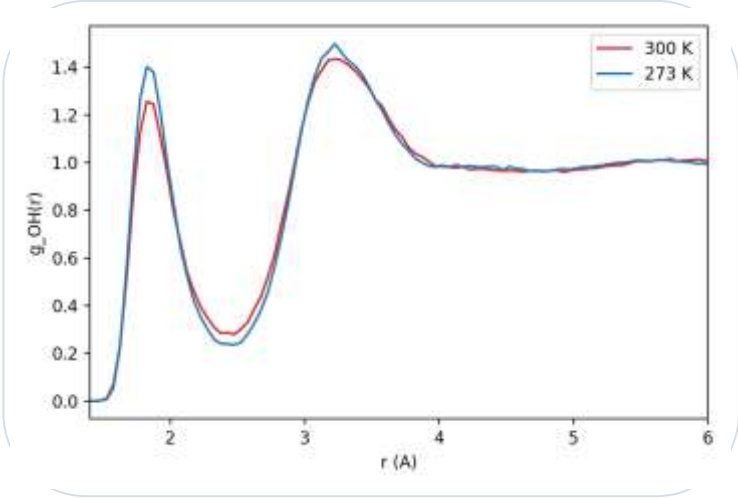
O–O



First peak
300 K: 2.775 Å, g=2.708
273 K: 2.775 Å, g=2.921

Same peak position, but a higher/sharper first shell at the cooler target.

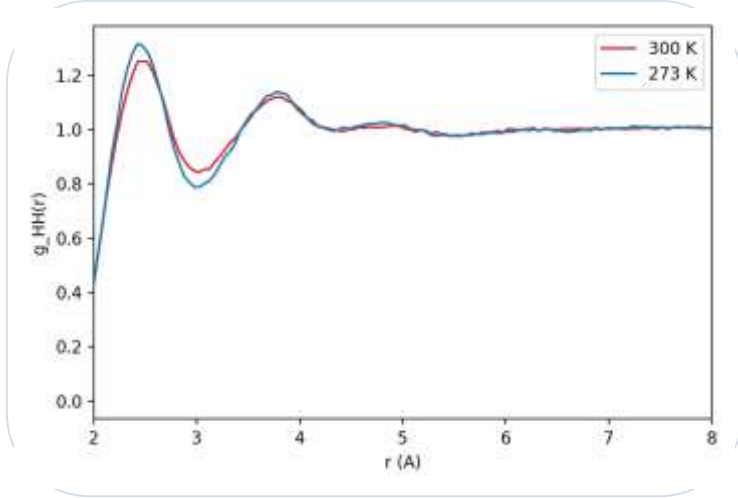
O–H



First feature: 1.825 Å
g: 1.256 → 1.401
Later robust feature: 3.225 Å

Short O–H feature: close intermolecular/H-bond-related contacts; later feature: further orientational/shell structure.

H–H



First feature: 2.475 Å → 2.425 Å
Later robust feature: 3.775 Å in both

The cooler branch shows stronger local H–H structural organization.

The Cooler Liquid Became More Tetrahedrally Ordered

q: 0.5617 → 0.5908

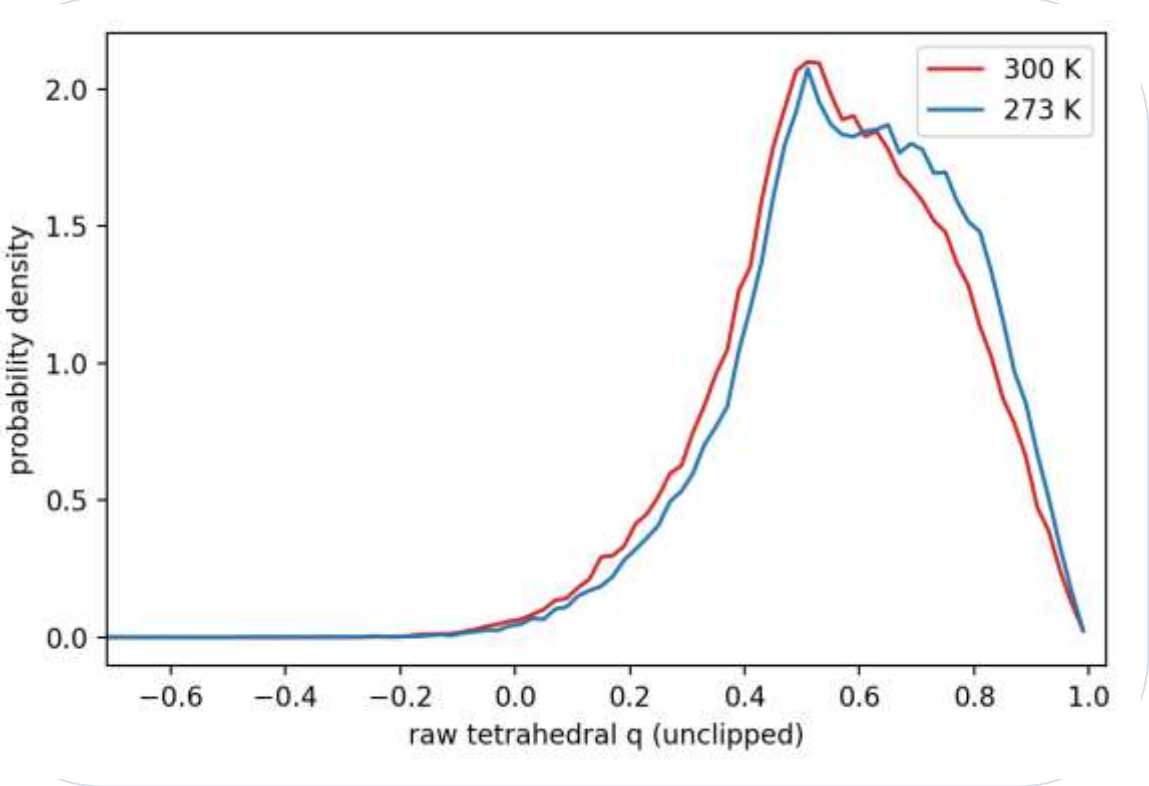
Higher q = stronger tetrahedral local ordering

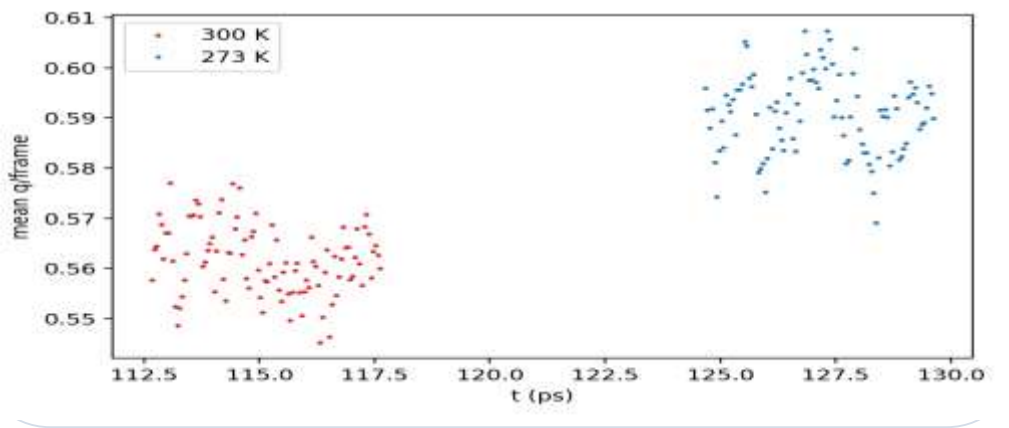
Same 3.5 Å cutoff

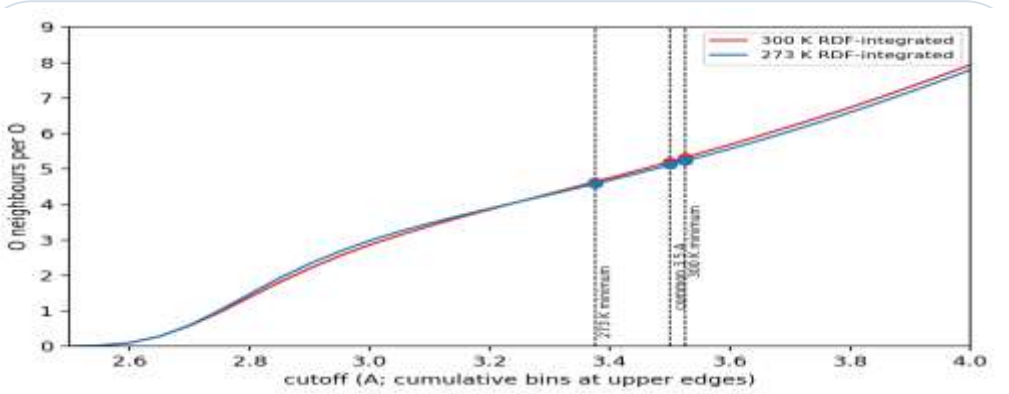
300 K: 5.2066 | 273 K: 5.1170 neighbors/O

Coordination changed only slightly at a matched radius.

We do not claim that cooling removed ~1 whole neighbor; that earlier large difference came from different cutoff radii.







Molecular Mobility: Mean Squared Displacement

At 1.0 ps
3.414 Å² (300 K) vs 2.380 Å² (273 K)

At 2.0 ps
6.436 Å² (300 K) vs 4.345 Å² (273 K)

The 300 K curve stays above the 273 K curve, so the warmer liquid has higher short-time oxygen mobility.



300 K shows consistently higher short-time oxygen mobility.
No common fit window met all criteria for a defensible diffusion coefficient.

Diffusion from MSD

For 3D diffusion: $D = (1/6) \cdot d(MSD)/dt$

So the diffusion coefficient is obtained from the slope of the approximately linear MSD region divided by 6.

Apparent short-time estimates:

- 300 K: $D \approx 5.17 \times 10^{-5}$ cm²/s
- 273 K: $D \approx 3.29 \times 10^{-5}$ cm²/s

Interpretation: the larger D at 300 K means the warmer liquid has higher molecular mobility.

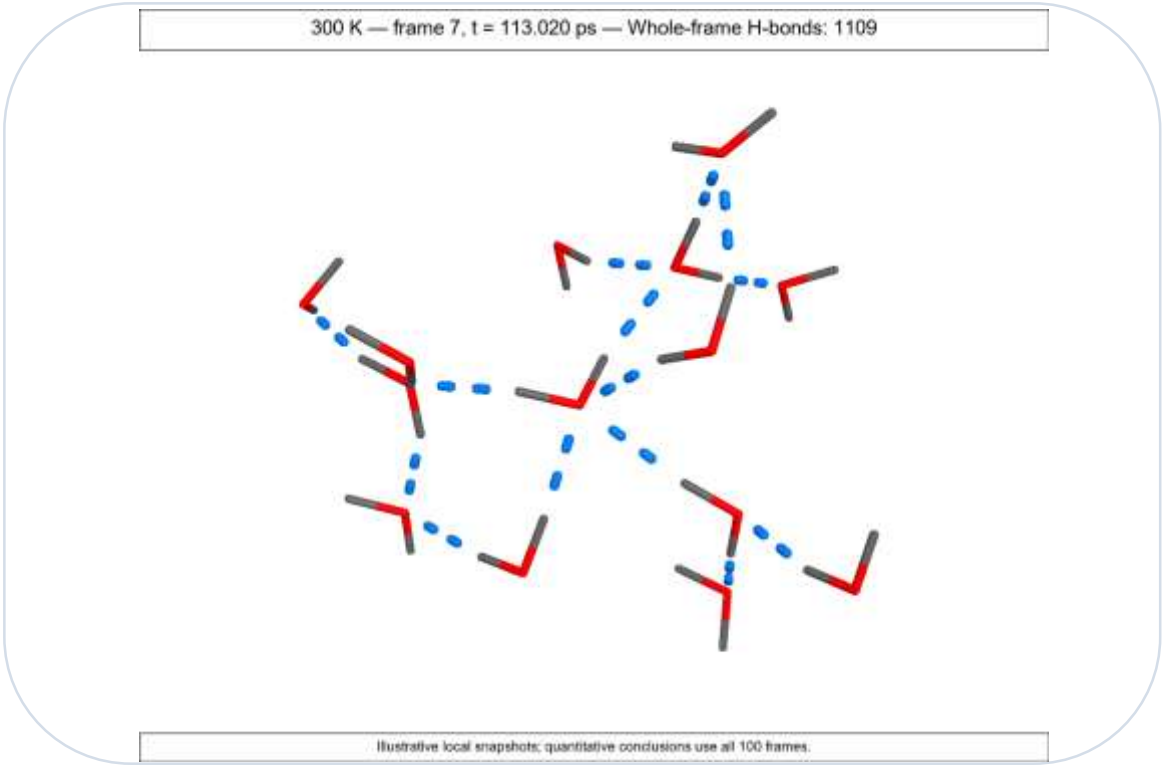
Note: these are apparent short-time estimates, not converged bulk diffusion coefficients, because the trajectories are only ~5 ps.

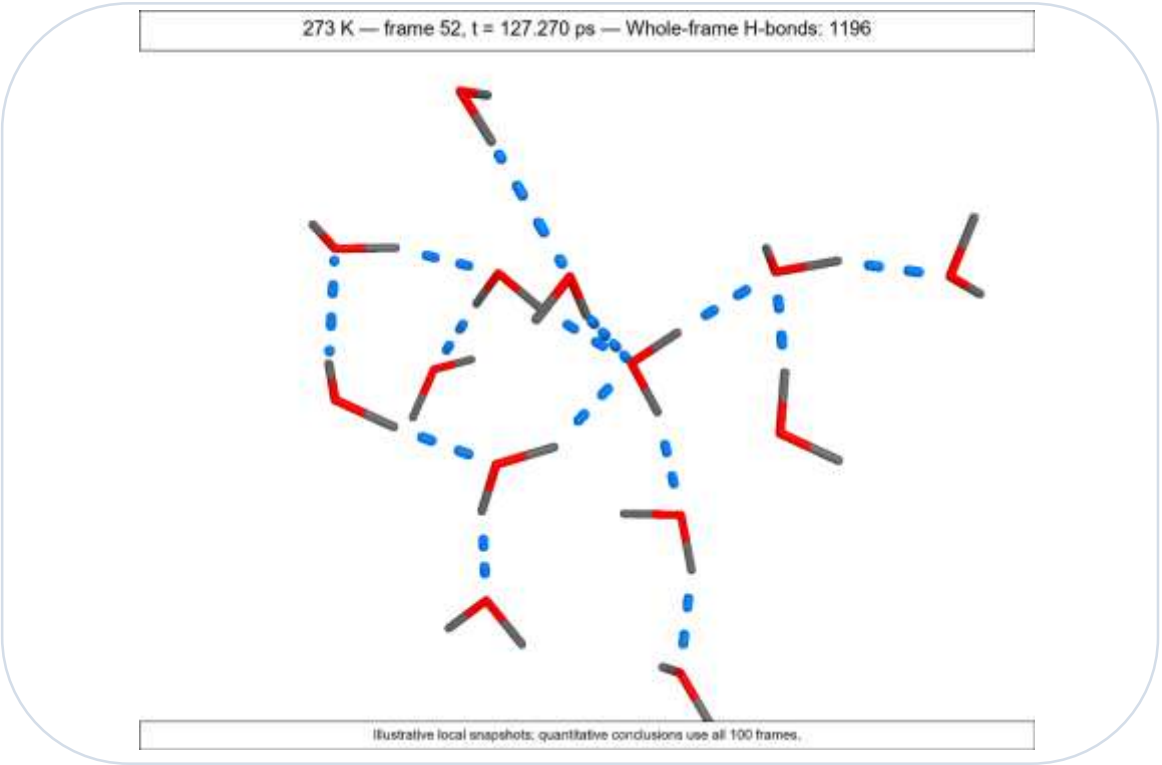
Hydrogen-Bond Network: Representative Local Snapshots

300 K representative frame
frame 7 • 113.020 ps • 1109 whole-frame H-bonds

273 K representative frame
frame 52 • 127.270 ps • 1196 whole-frame H-bonds

Quantitative result: the cooler trajectory has more geometric H-bonds overall (2.775 vs 2.572 H-bonds/water).





Illustrative local snapshots only; quantitative conclusions come from the full 100-frame analysis.

Putting All the Evidence Together

Thermodynamics

- lower sampled temperature
- lower potential energy
- very similar density/volume

Hydrogen-bond network

- more H-bond events
- slightly shorter O···O
- slightly straighter angles

Local structure

- sharper OO/OH features
- increased tetrahedral q

Coordination

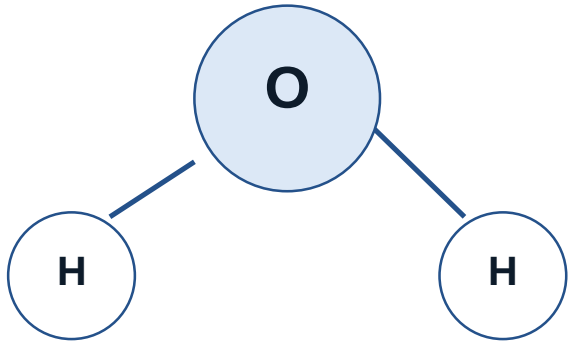
- only a small matched-radius change

Lowering the target temperature produced a more populated and more tetrahedrally ordered geometric hydrogen-bond network in this short liquid-water trajectory.

This is a structural/geometric conclusion—not direct proof of stronger H-bond energy.

Thank you

Questions?



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Molecular Dynamics